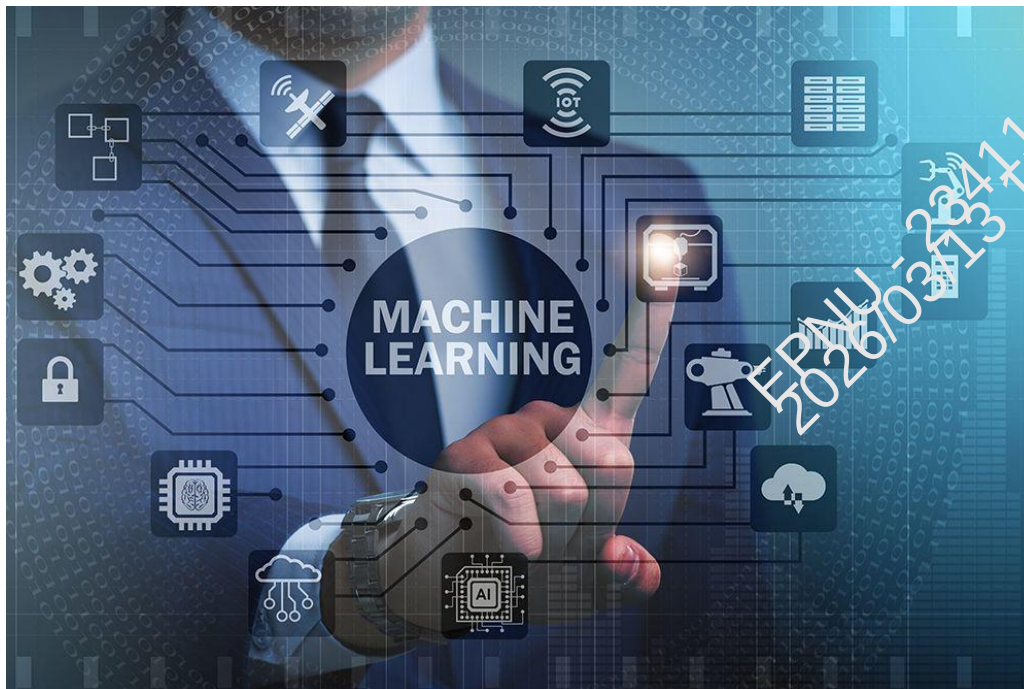


Machine Learning



Port said East University

Faculty of Computer Science and Information Technology



Machine Learning

Prepared by:

Dr. Sara Abo Hashish

Assistant Professor at Information Technology Department

Faculty of Management Technology&

Information Systems, Port Said University

Machine Learning

Contents

Chapter 1: Machine Learning	5
1. Definition of Machine Learning.....	6
2. Differences Between Machine Learning Algorithms and Traditional Rule-Based Algorithms.....	7
3. How do machines learn.....	9
4. Rational Understanding of Machine Learning Algorithms	10
5. Dataset.....	11
6. Data Processing	13
7. Machine Learning Classification.....	18
8. SUMMARY.....	20
9. Assignments chapter 1.....	23
10. SHORT-ANSWER TYPE QUESTIONS.....	19
	24
Chapter 2: Modelling and Evaluation	25
1. Predictive models.....	26
2. Descriptive models.....	26
3. Underfitting.....	27
4. Overfitting.....	29
5. Bias – variance trade-off.....	30
6. Evaluation Metrics.....	35
7. Summary...	38
8. Assignments chapter 2.....	40
9. SHORT-ANSWER TYPE QUESTIONS.....	
Chapter 3: Supervised Learning: Classification	
1. K Nearest Neighbors	41
2. Decision Tree.....	46
3. Random Forest Model.....	64
4. SVM.....	66
5. Summary.....	72
6. Assignments chapter 3.....	79
7. SHORT-ANSWER TYPE QUESTIONS.....	

Machine Learning

Chapter 4: Bayesian Concept Learning	80
1. Naive Bayes Classification.....	86
2. Summary.....	87
3. Assignments chapter 4.....	
 Chapter 5: Supervised Learning: Regression	88
1. Introduction	89
2. Common Algorithms.....	89
2.1 Simple Linear Regression	91
2.1.1 Slope of the simple linear regression model	97
2.1.2 No Relation.....	98
2.1.3 Error in simple regression.....	104
2.2 Multiple Linear Regression.....	107
3. Assignments chapter 5.....	111
4. SHORT ANSWER-TYPE QUESTIONS	
 Chapter 6: Unsupervised Learning	112
1. Clustering	121
2. Summary.....	122
3. Assignments chapter 6.....	125
4. SHORT-ANSWER TYPE QUESTIONS.....	
References	

Machine Learning

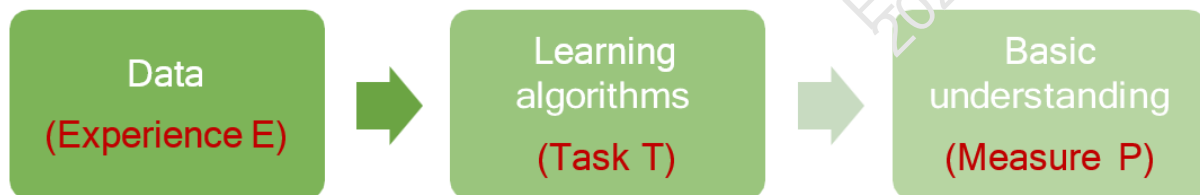
Chapter one

Machine Learning

Machine learning is a core research field of AI, and it is also a necessary knowledge for deep learning. Therefore, this chapter mainly introduces the main concepts of machine learning, the classification of machine learning, the overall process of machine learning, and the common algorithms of machine learning.

1. Definition of Machine Learning

Machine learning (including deep learning) is a study of learning algorithms. A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T , as measured by P , improves with experience E .



What this essentially means is that a machine can be considered to learn if it is able to gather experience by doing a certain task and improve its performance in doing the similar tasks in the future. When we talk about past experience, it means past data related to the task. This data is an input to the machine from some source. In the context of the learning to play checkers, E represents the experience of playing the game, T represents the task of playing checkers and P is the performance measure indicated by the percentage of games won by the player. The same mapping can be applied for any other machine learning problem, for example, image classification problem. In context of image classification, E represents the past data with images having labels or assigned classes (for example whether the image is of a class cat or a class dog or a class elephant etc.), T is the task of assigning class to new, unlabelled

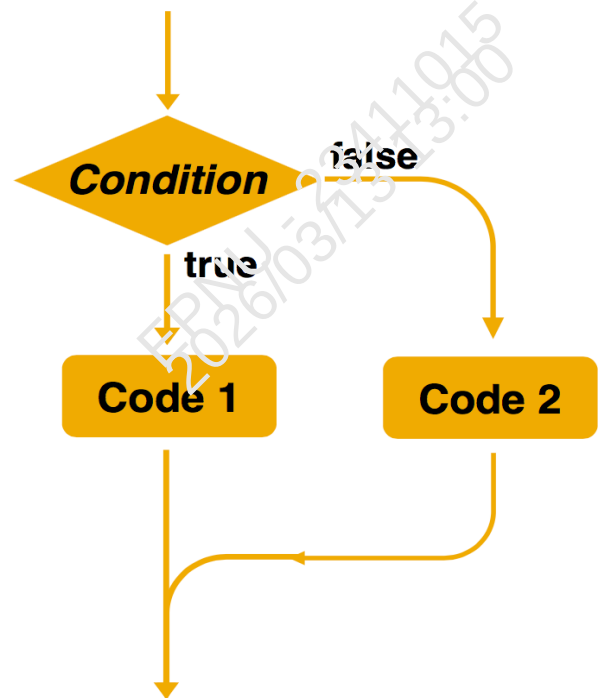
Machine Learning

images and P is the performance measure indicated by the percentage of images correctly classified

2. Differences Between Machine Learning Algorithms and Traditional Rule-Based Algorithms

Rule-based algorithms

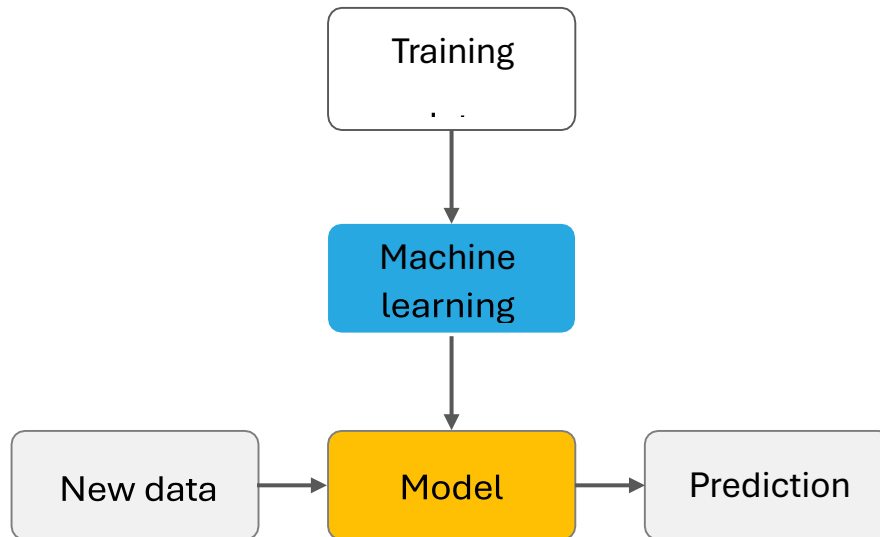
- Explicit programming is used to solve problems.
- Rules can be manually specified



Machine learning

- Samples are used for training.
- The decision-making rules are complex or difficult to describe.
- Rules are automatically learned by machines

Machine Learning



3. How do machines learn

The basic machine learning process can be divided into three parts.

1. Data Input: Past data or information is utilized as a basis for future decision-making
2. Abstraction: The input data is represented in a broader way through the underlying algorithm
3. Generalization: Generalization is the most important goal of machine learning. After learning patterns from training data, the model must apply this knowledge to new, unseen data.



Machine Learning

Let's consider the situation of typical process of learning from classroom and books and preparing for the examination. It is a tendency of many students to try and memorize (we often call it 'learn by heart') as many things as possible. This may work well when the scope of learning is not so vast. Also, the kinds of questions which are asked in the examination are pretty much simple and straightforward. The questions can be answered by simply writing the same things which have been memorized.

So, what we see in the case of human learning is that just by great memorizing and perfect recall, i.e. just based on knowledge input, students can do well in the examinations only till a certain stage. Beyond that, a better learning strategy needs to be adopted:

1. to be able to deal with the vastness of the subject matter and the related issues in memorizing it
2. to be able to answer questions where a direct answer has not been learnt

A good option is to figure out the key points or ideas amongst a vast pool of knowledge. This helps in creating an outline of topics and a conceptual mapping of those outlined topics with the entire knowledge pool.

For example, a broad pool of knowledge may consist of all living animals and their characteristics such as whether they live in land or water, whether they lay eggs, whether they have scales or fur or none, etc. It is a difficult task for any student to memorize the characteristics of all living animals – no matter how much photographic memory he/she may possess. It is better to draw a notion about the basic groups that all living animals belong to and the characteristics which define each of the basic groups. The basic groups of animals are invertebrates and vertebrates. Vertebrates are

Machine Learning

further grouped as mammals, reptiles, amphibians, fishes, and birds. Here, we have mapped animal groups and their salient characteristics.

1. Invertebrate: Do not have backbones and skeletons
2. Vertebrate
 1. Fishes: Always live in water and lay eggs
 2. Amphibians: Semi-aquatic i.e. may live in water or land; smooth skin; lay eggs
 3. Reptiles: Semi-aquatic like amphibians; scaly skin; lay eggs; cold-blooded
 4. Birds: Can fly; lay eggs; warm-blooded
 5. Mammals: Have hair or fur; have milk to feed their young; warm-blooded

During the machine learning process, knowledge is fed in the form of input data. However, the data cannot be used in the original shape and form. As we saw in the example above, abstraction helps in deriving a conceptual map based on the input data. This map, or a **model** as it is known in the machine learning paradigm, is summarized knowledge representation of the raw data.

The **model** may be in any one of the following forms

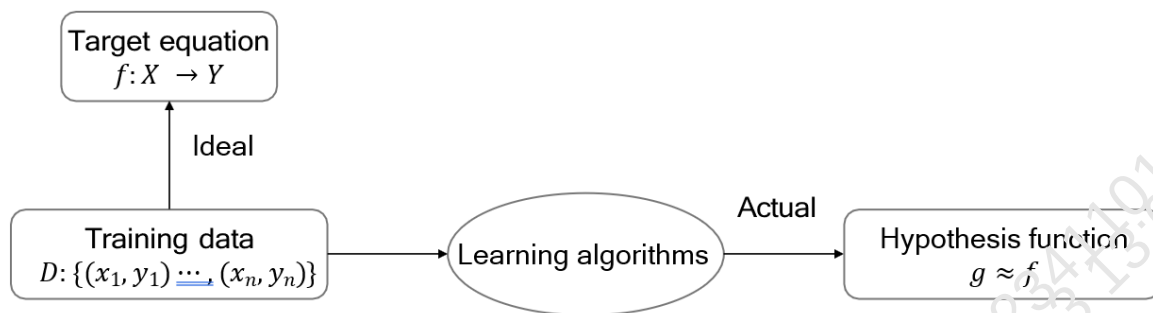
- Computational blocks like if/else rules
- Mathematical equations
- Specific data structures like trees or graphs
- Logical groupings of similar observation

The first part of machine learning process is **abstraction** i.e. abstract the knowledge which comes as input data in the form of a model. However, this abstraction process, or more popularly training the model, is just one part of machine learning. The other key part is to tune up the **abstracted** knowledge to a form which can be used to take

Machine Learning

future decisions. This is achieved as a part of generalization. This part is quite difficult to achieve. This is because the model is trained based on a finite set of data, which may possess a limited set of characteristics.

4. Rational Understanding of Machine Learning Algorithms



- Target function f is unknown. Learning algorithms cannot obtain a perfect function f .
- Assume that hypothesis function g **approximates** function f , but may be different from function f .

5. Dataset

Dataset: a collection of data used in machine learning tasks. Each data record is called a sample. Events or attributes that reflect the performance or nature of a sample in a particular aspect are called features.

Training set: a dataset used in the training process, where each sample is referred to as a training sample. The process of creating a model from data is called learning (training).

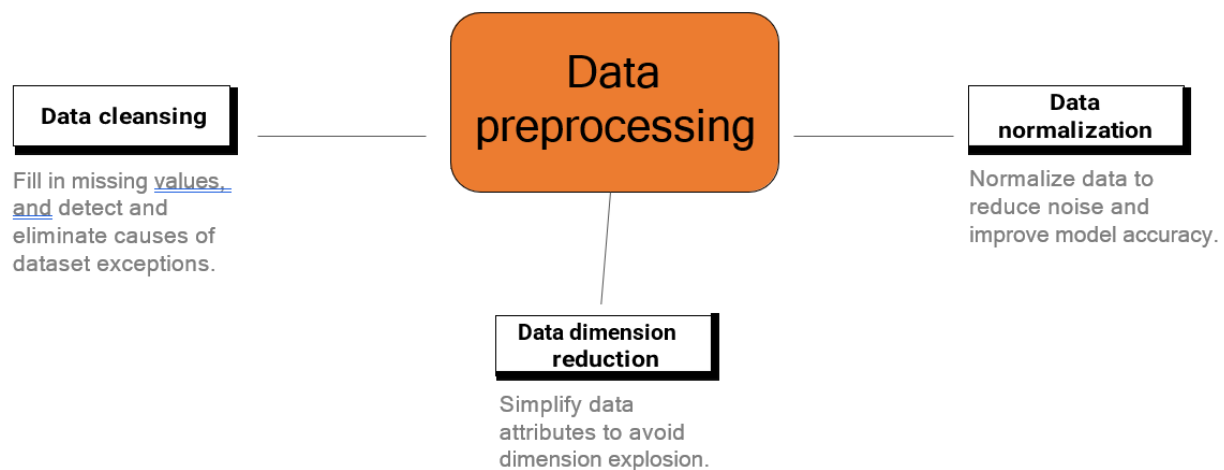
Test set: Testing refers to the process of using the model obtained after learning for prediction. The dataset used is called a test set, and each sample is called a test sample.

Machine Learning

		Feature 1	Feature 2	Feature 3	Label	
		No.	Area	School Districts	Direction	House Price
Training set	1	100	8	South	1000	
	2	120	9	Southwest	1300	
	3	60	6	North	700	
	4	80	9	Southeast	1100	
Test set	5	95	3	South	850	

6. Data Processing

Data is crucial to models. It is the ceiling of model capabilities. Without good data, there is no good model.



Machine Learning

Data Cleansing

Most machine learning models process features, which are usually numeric representations of input variables that can be used in the model. In most cases, the collected data can be used by algorithms only after being preprocessed.

The preprocessing operations include the following:

- ❑ Data filtering
- ❑ Processing of lost data
- ❑ Processing of possible exceptions, errors, or abnormal values
- ❑ Data consolidation (Combination of data from multiple data source)

Generally, real data may have some quality problems.

- ❑ Incompleteness: contains missing values or the data that lacks attributes
- ❑ Noise: contains incorrect records or exceptions.
- ❑ Inconsistency: contains inconsistent records.

Data Conversion

After being preprocessed, the data needs to be converted into a representation form suitable for the machine learning model. Common data conversion forms include the following:

- ❑ With respect to classification, category data is encoded into a corresponding numerical representation.
- ❑ Value data is converted to category data to reduce the value of variables (for age segmentation).
- ❑ Other data

Machine Learning

- In the text, the word is converted into a word vector through word embedding (generally using the word2vec model, BERT model, etc).
- Process image data (color space, grayscale, geometric change, Haar feature, and image enhancement)
- Feature engineering
 - Normalize features to ensure the same value ranges for input variables of the same model.
 - Feature expansion: Combine or convert existing variables to generate new features, such as the average.

7. Machine Learning Classification

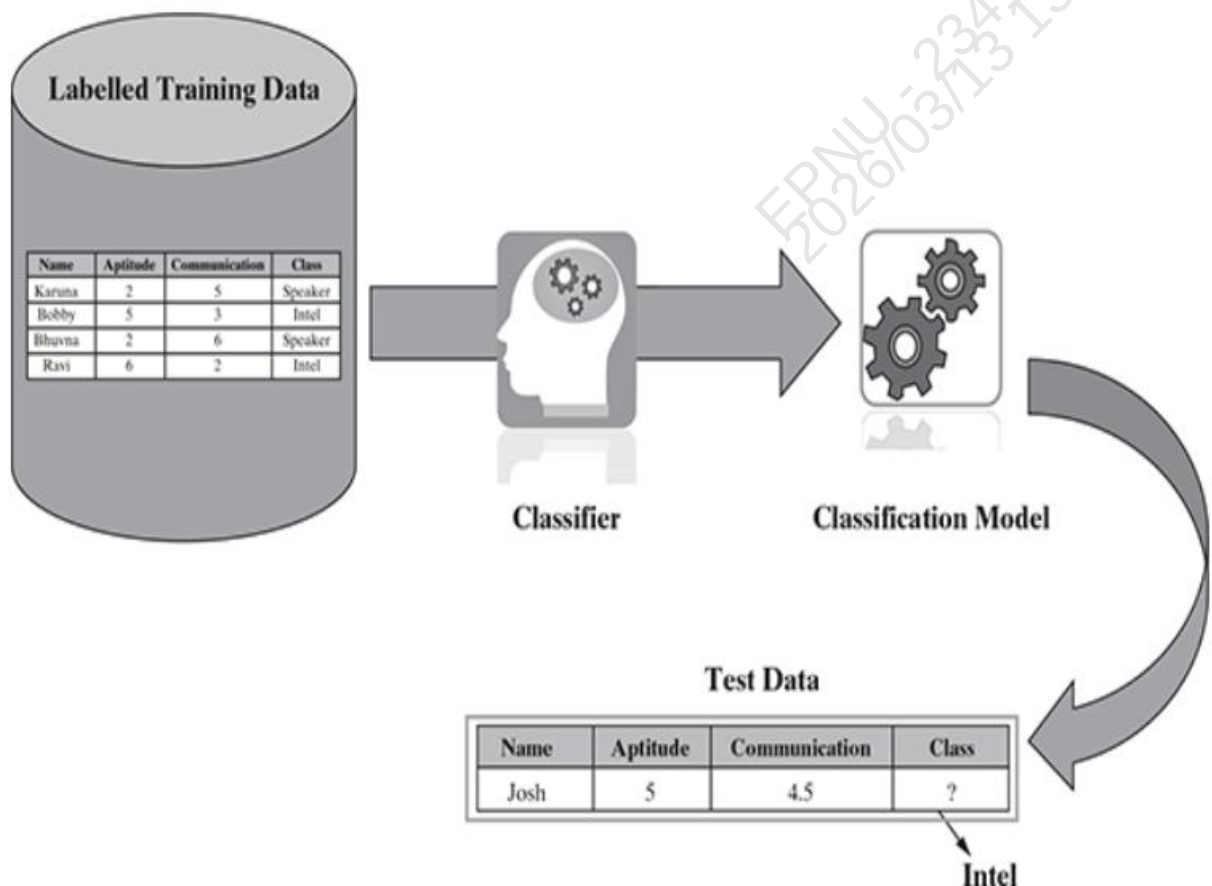
- **Supervised learning:** Obtain an optimal model with required performance through training and learning based on the samples of known categories. Then, use the model to map all inputs to outputs and check the output for the purpose of classifying unknown data. Supervised learning can be further classified into two types – Regression and Classification.

Classification: A computer program needs to specify which of the k categories some input belongs to. To accomplish this task, learning algorithms usually output a function $f: R^n \rightarrow (1, 2, \dots, k)$. such as analyzing positive/negative sentiment, male and female persons, benign and malignant tumors, secure and unsecure loans etc.

Machine Learning

Some typical classification problems include:

- ✚ Image classification
- ✚ Prediction of disease
- ✚ Win-loss prediction of games
- ✚ Prediction of natural calamity like earthquake, flood, etc. Recognition of handwriting



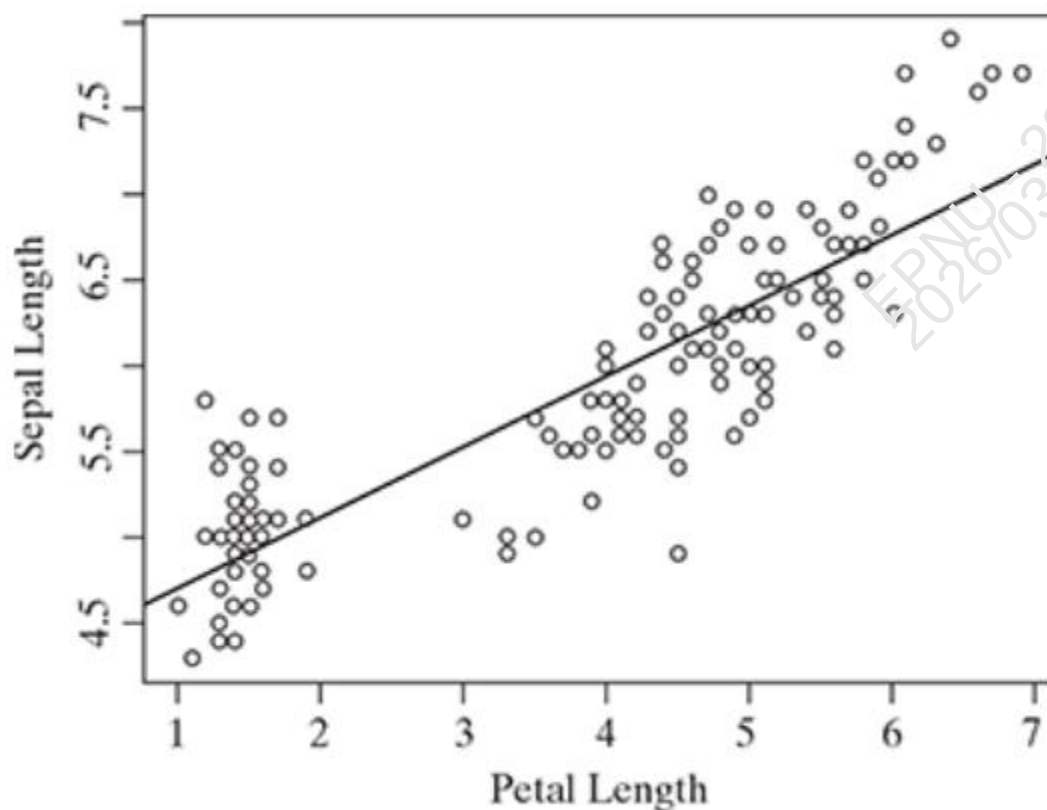
Regression: For this type of task, a computer program predicts the output for the given input. Learning algorithms typically output a function $f: R^n \rightarrow R$. An example predicting real estate prices.

Machine Learning

Figure shows a typical simple regression model, where regression line is fitted based on values of target variable with respect to different values of predictor variable. A typical linear regression model can be represented in the form

$$y = \alpha + \beta x$$

where 'x' is the predictor variable and 'y' is the target variable.



Typical applications of regression can be seen in

- ✚ Demand forecasting in retails
- ✚ Sales prediction for managers
- ✚ Price prediction in real estate
- ✚ Weather forecast
- ✚ Skill demand forecast in job market

Machine Learning

In supervised learning, learning data comes with description, labels, targets or desired outputs and the objective is to find a general rule that maps inputs to outputs. This kind of learning data is called **labeled data**. The learned rule is then used to label new data with unknown outputs.

Supervised learning involves building a machine learning model that is based on **labeled samples**. For example, if we build a system to estimate the price of a plot of land or a house based on various features, such as size, location, and so on, we first need to create a database and label it. We need to teach the algorithm what features correspond to what prices. Based on this data, the algorithm will learn how to calculate the price of real estate using the values of the input features.

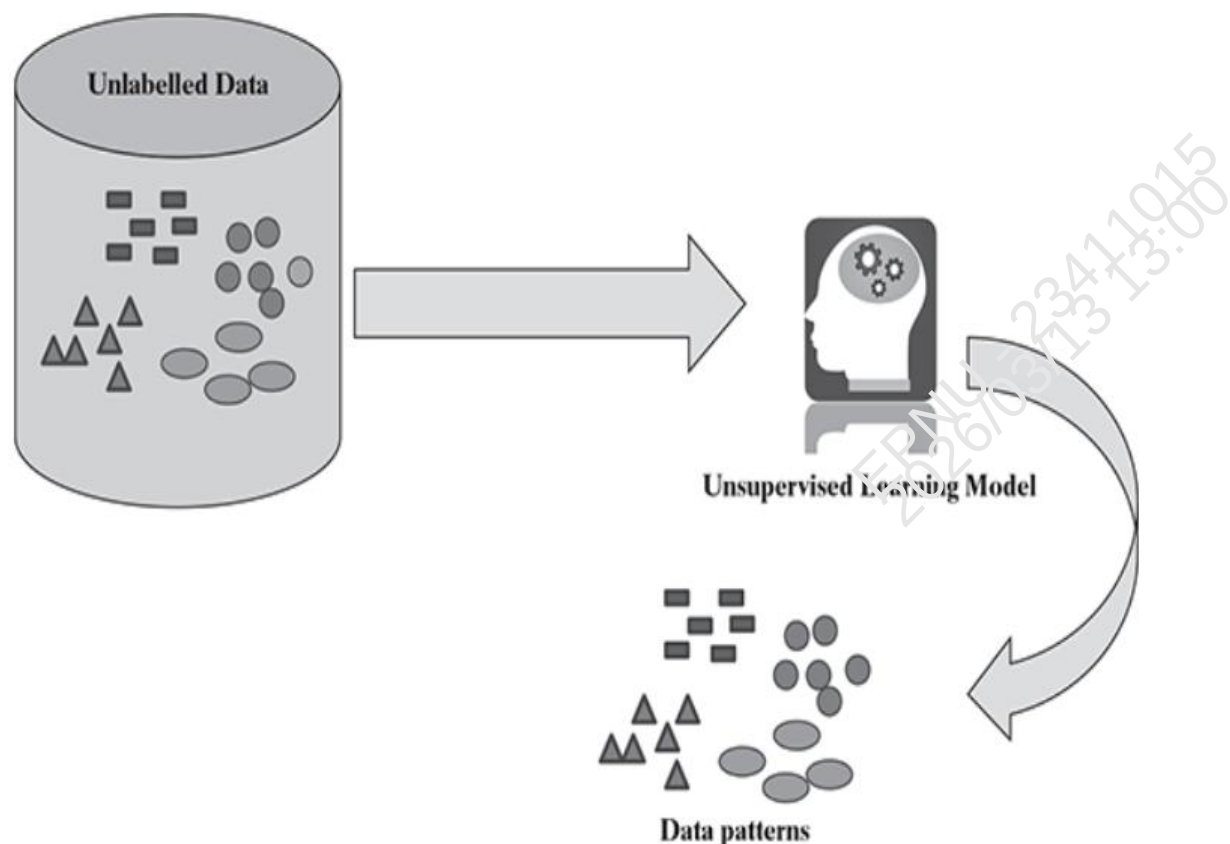
Supervised learning deals with learning a function from available training data. Here, a learning algorithm analyzes the training data and produces a derived function that can be used for mapping new examples. There are many **supervised learning algorithms** such as Logistic Regression, Neural networks, Support Vector Machines (SVMs), and Naive Bayes classifiers.

Common **examples** of supervised learning include classifying e-mails into spam and not-spam categories, labeling webpages based on their content, and voice recognition.

- **Unsupervised learning:** For **unlabeled** samples, the learning algorithms directly model the input datasets. Clustering is a common form of unsupervised learning. We only need to put highly similar samples together, calculate the similarity between new samples and existing ones, and classify them by similarity.

Machine Learning

Clustering: A large amount of data from an unlabeled dataset is divided into multiple categories according to internal similarity of the data. Data in the same category is more similar than that in different categories. This feature can be used in scenarios such as image retrieval and user profile management.

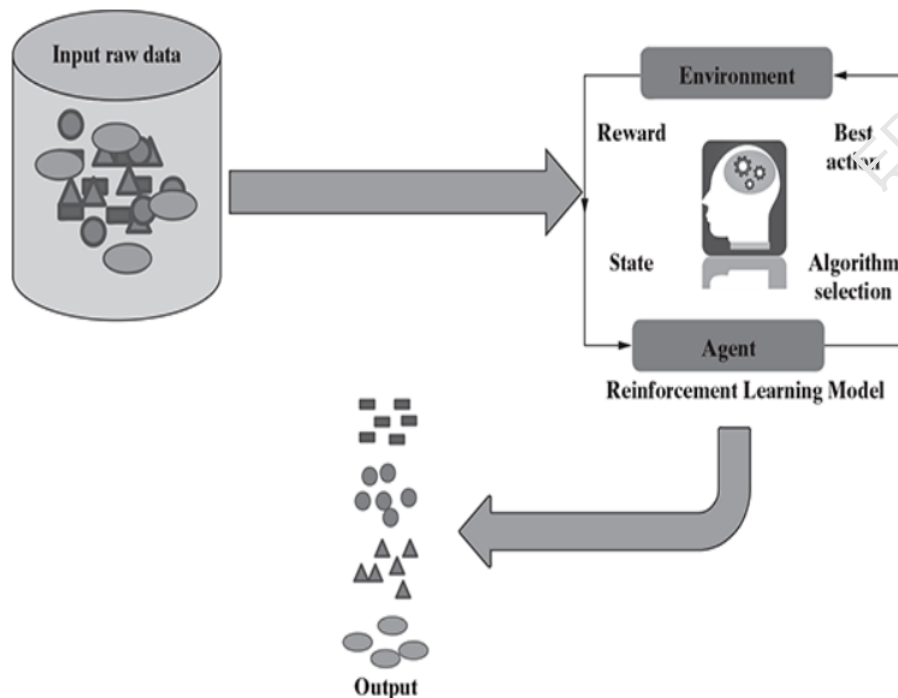


- **Semi-supervised learning:** If some learning samples are labeled, but some other are not labeled, then it is semi-supervised learning. It makes use of a large amount of **unlabeled data for training** and a small amount of **labeled data for testing**. Semi-supervised learning is applied in cases where it is expensive to acquire a fully labeled dataset while more practical to label a small subset. For example, it often requires skilled experts to label certain remote sensing images, and lots of field

Machine Learning

experiments to locate oil at a particular location, while acquiring unlabeled data is relatively easy.

- **Reinforcement learning:** It is an area of machine learning concerned with how agents ought to take actions in an environment to maximize some notion of cumulative reward. The difference between reinforcement learning and supervised learning is the teacher signal. The reinforcement signal provided by the environment in reinforcement learning is used to evaluate the action (scalar signal) rather than telling the learning system how to perform correct actions.



Machine Learning

8. SUMMARY

- + Machine learning imbibes the philosophy of human learning, i.e. learning from expert guidance and from experience.
- + The basic machine learning process can be divided into three parts.
 - Data Input: Past data or information is utilized as a basis for future decision-making.
 - Abstraction: The input data is represented in a summarized way
 - Generalization: The abstracted representation is generalized to form a framework for making decisions.
- + Before starting to solve any problem using machine learning, it should be decided whether the problem is a right problem to be solved using machine learning.
- + Machine learning can be classified into three broad categories:
 - Supervised learning: Also called predictive learning. The objective of this learning is to predict class/value of unknown objects based on prior information of similar objects. Examples: predicting whether a tumour is malignant or benign, price prediction in domains such as real estate, stocks, etc.
 - Unsupervised learning: Also called descriptive learning, helps in finding groups or patterns in unknown objects by grouping similar objects together. Examples: customer segmentation, recommender systems, etc.
 - Reinforcement learning: A machine learns to act on its own to achieve the given goals. Examples: self-driving cars, intelligent robots, etc.
- + Machine learning has been adopted by various industry domains such as Banking and Financial Services, Insurance, Healthcare, Life Sciences, etc. to solve problems.

Machine Learning

9. Assignments chapter 1

1. Machine learning is ____ field.

1. Inter-disciplinary
2. Single
3. Multi-disciplinary
4. All of the above

2. A computer program is said to learn from _____ E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with E.

1. Training
2. Experience
3. Database
4. Algorithm

3. _____ has been used to train vehicles to steer correctly and autonomously on road.

1. Machine learning
2. Data mining
3. Neural networks
4. Robotics

4. Any hypothesis find an approximation of the target function over a sufficiently large set of training examples will also approximate the target function well over other unobserved examples. This is called ____.

1. Hypothesis
2. Inductive hypothesis
3. Learning
4. Concept learning

Machine Learning

5. Factors which affect performance of a learner system does not include

1. Representation scheme used
2. Training scenario
3. Type of feedback
4. Good data structures

6. Different learning methods does not include

1. Memorization
2. Analogy
3. Deduction
4. Introduction

7. A model of language consists of the categories which does not include

1. Language units
2. Role structure of units
3. System constraints
4. Structural units

8. How many types are available in machine learning?

1. 1
2. 2
3. 3
4. 4

9. The k -means algorithm is a

1. Supervised learning algorithm
2. Unsupervised learning algorithm
3. Semi-supervised learning algorithm

4. Weakly supervised learning algorithm

Machine Learning

10. The Q-learning algorithm is a

1. Supervised learning algorithm
2. Unsupervised learning algorithm
3. Semi-supervised learning algorithm
4. Reinforcement learning algorithm

11. This type of learning to be used when there is no idea about the class or label of a particular data

1. Supervised learning algorithm
2. Unsupervised learning algorithm
3. Semi-supervised learning algorithm
4. Reinforcement learning algorithm

12. The model learns and updates itself through reward/punishment in case of

1. Supervised learning algorithm
2. Unsupervised learning algorithm
3. Semi-supervised learning algorithm
4. Reinforcement learning algorithm

Machine Learning

10.SHORT-ANSWER TYPE QUESTIONS

1. What is classification? Explain the key differences between classification and regression.
2. What is regression? Give example of some practical problems solved using regression.
3. Explain the process of clustering in details.
4. Write short notes on any two of the following:
 - Application of machine learning algorithms
 - Supervised learning
 - Unsupervised learning
 - Reinforcement learning

EPNU - 23411015
2026/03/13 13:00

Machine Learning

Chapter Two

Modelling and Evaluation

In machine learning paradigm, the potential causes of disturbance, e.g. average income of the local population, weapon sales, the inflow of immigrants, etc. are input variables. They are also called predictors, attributes, features, independent variables, or simply variables. The number of criminal incidents is an **output variable** (also called **response or dependent variable**). **Input variables** can be denoted by X , while individual input variables are represented as $X_1, X_2, X_3, \dots, X_n$ and **output variable** by symbol Y .

The relationship between X and Y is represented in the general form:

$$Y = f(X) + e,$$

where ' f ' is the **target function** and ' e ' is a **random error term**.

10. Predictive models

Models for supervised learning or predictive models, as is understandable from the name itself, try to predict certain value using the values in an input data set. The learning model attempts to establish a relation between the target feature, i.e. the feature being predicted, and the predictor features. The predictive models have a clear focus on what they want to learn and how they want to learn.

Predictive models, in turn, may need to predict the value of a category or class to which a data instance belongs to. Below are some examples:

1. Predicting win/loss in a cricket match
2. Predicting whether a transaction is fraud
3. Predicting whether a customer may move to another product

Machine Learning

The models which are used for prediction of target features of categorical value are known as classification models. The target feature is known as a class and the categories to which classes are divided into are called levels.

Some of the popular classification models include ***k*-Nearest Neighbor (kNN)**, **Naïve Bayes**, and **Decision Tree**.

11.Descriptive models

Models for unsupervised learning or descriptive models are used to describe a data set or gain insight from a data set. There is no target feature or single feature of interest in case of unsupervised learning. Based on the value of all features, interesting patterns or insights are derived about the data set.

Descriptive models which group together similar data instances, i.e. data instances having a similar value of the different features are called clustering models. Examples of clustering include

1. Customer grouping or segmentation based on social, demographic, ethnic, etc. factors
2. Grouping of music based on different aspects like genre, language, timeperiod, etc.
3. Grouping of commodities in an inventory

The most popular model for clustering is *k*-Means

Machine Learning

12.

U

nderfitting

If the target function is kept too simple, it may not be able to capture the essential nuances and represent the underlying data well. A typical case of underfitting may occur when trying to represent a non-linear data with a linear model as demonstrated by both cases of underfitting shown in figure. **Underfitting:** occurs when the model or the algorithm does not fit the data well enough.

Many times underfitting happens due to unavailability of sufficient training data. Underfitting results in both poor performance with training data as well as poor generalization to test data. Underfitting can be avoided by

1. using more training data
2. reducing features by effective feature selection

13.

O

verfitting

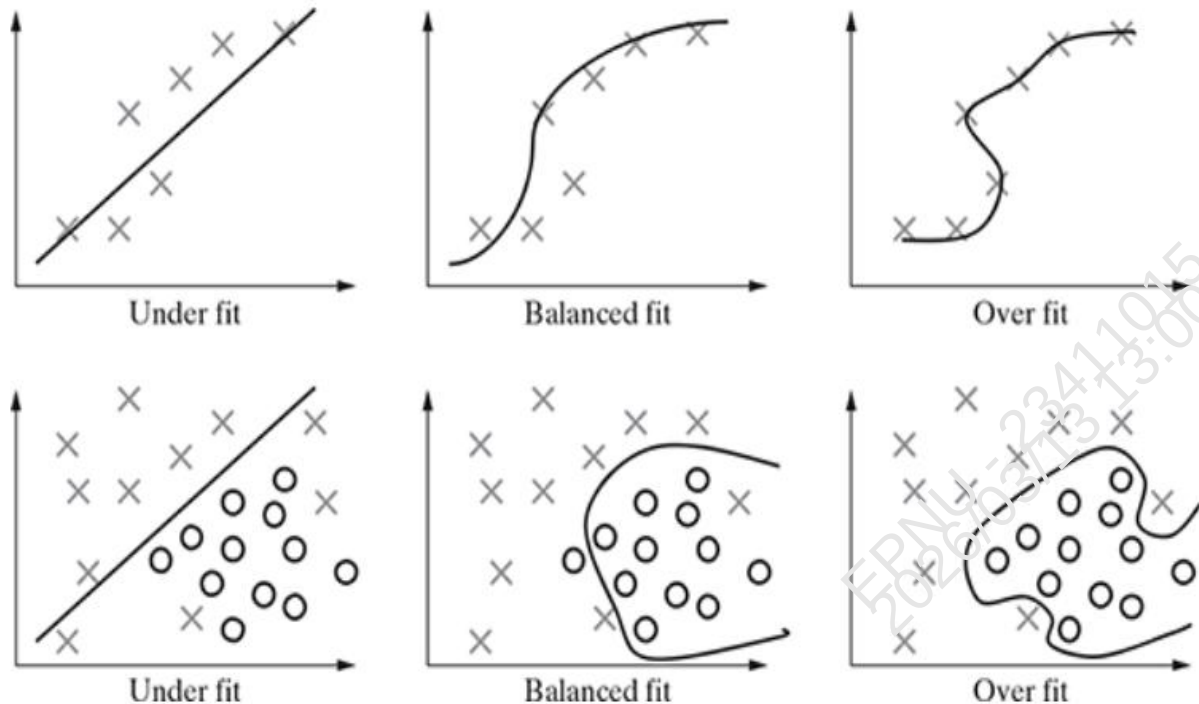
Overfitting occurs when the training error of the model obtained after learning is small but the generalization error is large (poor generalization capability).

Overfitting results in good performance with training data set, but poor generalization and hence poor performance with test data set. Overfitting can be avoided by:

1. using re-sampling techniques like k -fold cross validation
2. hold back of a validation data set
3. remove the nodes which have little or no predictive power for the given machine learning problem.

Machine Learning

Both underfitting and overfitting result in poor classification quality which is reflected by low classification accuracy



14.

B

bias – variance trade-off

In supervised learning, the class value assigned by the learning model built based on the training data may differ from the actual class value. This error in learning can be of two types

- errors due to 'bias'
- error due to 'variance'.

14.1 Errors due to 'Bias'

Errors due to bias arise from simplifying assumptions made by the model to make the target function less complex or easier to learn. In short, it is due to underfitting of the model. **Bias is Difference between the expected (or average) prediction value and the**

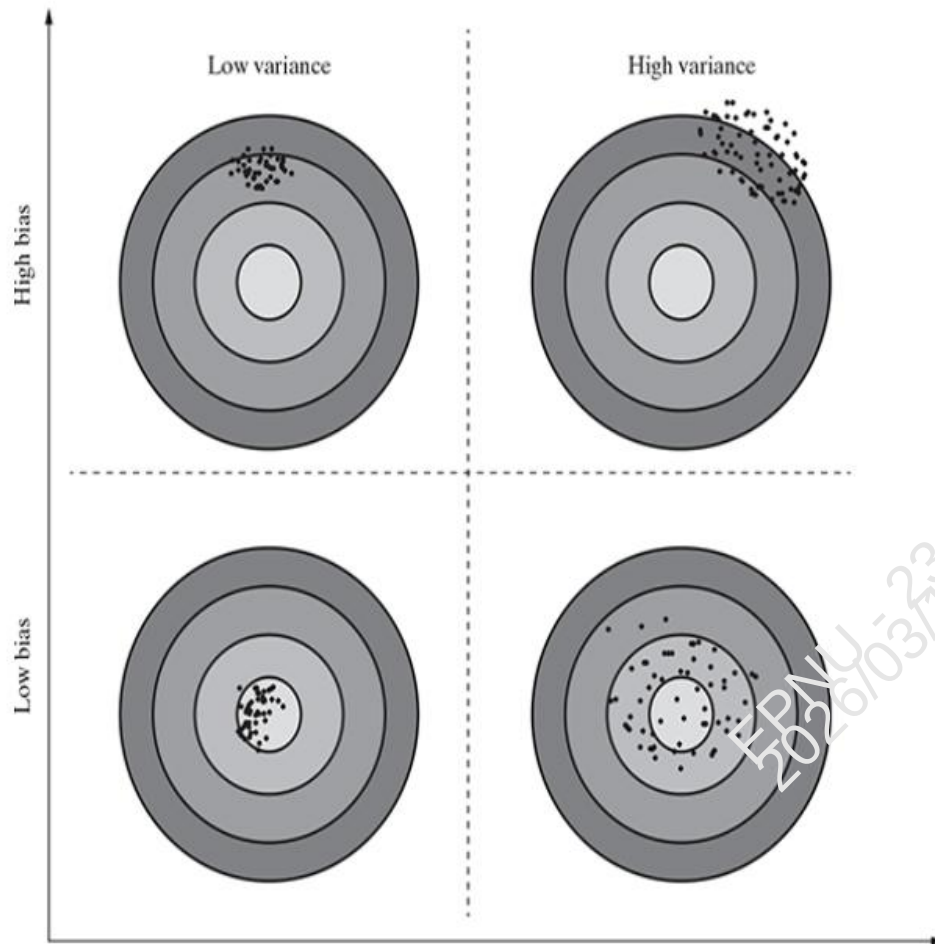
Machine Learning

correct value we are trying to predict. Parametric models generally have high bias making them easier to understand/interpret and faster to learn. These algorithms have a poor performance on data sets, which are complex in nature and do not align with the simplifying assumptions made by the algorithm. Underfitting results in high bias.

14.2 Errors due to ‘Variance’

Errors due to variance occur from difference in training data sets used to train the model. Different training data sets (randomly sampled from the input data set) are used to train the model. Ideally the difference in the data sets should not be significant and the model trained using different training data sets should not be too different. However, in case of overfitting, since the model closely matches the training data, even a small difference in training data gets magnified in the model.

Machine Learning



Combinations of variance and bias are as

- ❑ Low bias & low variance → Good model
- ❑ Low bias & high variance
- ❑ High bias & low variance
- ❑ High bias & high variance → Poor model

Machine Learning

On one hand, **parametric algorithms** are generally seen to demonstrate high bias but low variance. On the other hand, **non-parametric algorithms** demonstrate low bias and high variance.

15. Evaluation Metrics

A **confusion matrix** is a performance measurement tool used to evaluate the effectiveness of a classification algorithm. It is a table that allows us to visualize the performance of a machine learning model, specifically in terms of its ability to correctly classify instances in each category.

The confusion matrix is particularly useful in classification problems, where the goal is to categorize data into discrete classes. It compares the actual (true) labels with the predicted labels made by the model, providing insights into both the errors the model is making and where those errors occur.

The confusion matrix is typically structured as a 2x2 table for binary classification problems, but it can be extended to multi-class classification tasks.

For binary classification, the matrix looks like this:

	Predicted Positive (P)	Predicted Negative (N)
Actual Positive (P)	True Positive (TP)	False Negative (FN)
Actual Negative (N)	False Positive (FP)	True Negative (TN)

Machine Learning

Key Terms in a Confusion Matrix:

1. **True Positive (TP):** The number of instances correctly classified as positive.
 - Example: A model correctly predicting that an email is spam.
2. **True Negative (TN):** The number of instances correctly classified as negative.
 - Example: A model correctly predicting that an email is not spam.
3. **False Positive (FP):** The number of instances incorrectly classified as positive (Type I error).
 - Example: A model wrongly classifying a non-spam email as spam.
4. **False Negative (FN):** The number of instances incorrectly classified as negative (Type II error).
 - Example: A model wrongly classifying a spam email as not spam.

Evaluation Metrics Derived from the Confusion Matrix:

Several performance metrics can be derived from the confusion matrix to assess the model's effectiveness:

1. **Accuracy:** The proportion of correct predictions (both true positives and true negatives) out of all predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Machine Learning

- 2. Precision (Positive Predictive Value):** The proportion of positive predictions that are actually correct.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- 3. Recall (Sensitivity or True Positive Rate):** The proportion of actual positives that are correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- 4. F1-Score:** The harmonic mean of precision and recall, providing a balanced measure when there is an imbalance between the two.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- 5. Specificity (True Negative Rate):** The proportion of actual negatives that are correctly identified.

$$\text{Specificity} = \frac{TN}{TN + FP}$$

- 6. False Positive Rate:** The proportion of actual negatives that are incorrectly classified as positive.

$$\text{False Positive Rate} = \frac{FP}{TN + FP}$$

- 7. False Negative Rate:** The proportion of actual positives that are incorrectly classified as negative.

Machine Learning

$$\text{False Negative Rate} = \frac{FN}{TP + FN}$$

Example 1

Consider a model predicting whether an email is spam or not. The confusion matrix might look like this:

	Predicted Spam (P)	Predicted Not Spam (N)
Actual Spam (P)	90	10
Actual Not Spam (N)	5	95

- **True Positive (TP) = 90:** The model correctly identified 90 spam emails.
- **False Positive (FP) = 5:** The model wrongly classified 5 non-spam emails as spam.
- **False Negative (FN) = 10:** The model wrongly classified 10 spam emails as not spam.
- **True Negative (TN) = 95:** The model correctly identified 95 non-spam emails.

Example 2

We have trained a machine learning model to identify whether the object in an image is a cat. Now we use 200 pictures to verify the model performance. Among the 200 images, objects in 170 images are cats, while others are not. The identification result of the model is that objects in 160 images are cats, while others are not.

Machine Learning

Estimated amount Actual amount	<i>yes</i>	<i>no</i>	Total
<i>yes</i>	140	30	170
<i>no</i>	20	10	30
Total	160	40	200

Precision: $P = \frac{TP}{TP+FP} = \frac{140}{140+20} = 87.5\%$

Recall: $R = \frac{TP}{P} = \frac{140}{170} = 82.4\%$

Accuracy: $ACC = \frac{TP+TN}{P+N} = \frac{140+10}{170+30} = 75\%$

Example 3:

	ACTUAL WIN	ACTUAL LOSS
Predicted Win	85	4
Predicted Loss	2	9

In context of the above confusion matrix, total count of TPs = 85, count of FPs = 4, count of FNs = 2 and count of TNs = 9

Machine Learning

$$\therefore \text{Model accuracy} = \frac{TP + TN}{TP + FP + FN + TN} = \frac{85 + 9}{85 + 4 + 2 + 9} = \frac{94}{100} = 94\%$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{85}{85 + 2} = \frac{85}{87} = 97.7\%$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{85}{85 + 4} = \frac{85}{89} = 95.5\%$$

EPNU - 23411015
2026/03/13 13:00

Machine Learning

8. Summary

- Target function of a model is the function defining the relationship between the input (also called predictor or independent) variables and the output (also called response or dependent or target) variable.

It is represented in the general form: $Y = f(X) + e$, where Y is the output variable, X represents the input variables and 'e' is a random error term.

- **Overfitting** refers to a situation where the model has been designed in such a way that it emulates the training data too closely. In such a case, any specific nuance in the training data, like noise or outliers, gets embedded in the model. It adversely impacts the performance of the model on the test data. In supervised learning, the value predicted by the learning model built based on the training data may differ from the actual class value. **This error in learning can be of two types – errors due to 'bias' and error due to 'variance'**. Errors due to bias arise from simplifying assumptions made by the model whereas errors due to variance occur from over-aligning the model with the training data sets.

- **For any classification model**, model accuracy is the primary indicator of the goodness of the model. It is given by a total number of correct classifications (either as the class of interest, or as not the class of interest) divided by total number of classifications done. **There are other indicators like error rate, sensitivity, specificity, precision and recall.**

- **Confusion matrix** is a performance measurement tool used to evaluate the effectiveness of a classification algorithm. It is a table that allows us to visualize the performance of a machine learning model, specifically in terms of its ability to correctly classify instances in each category.

Machine Learning

Terms and definitions:

- P : positive, indicating the number of real positive cases in the data.
- N : negative, indicating the number of real negative cases in the data.
- TP : true positive, indicating the number of positive cases that are correctly classified by the classifier.
- TP : true positive, indicating the number of positive cases that are correctly classified by the classifier.
- TN : true negative, indicating the number of negative cases that are correctly classified by the classifier.
- FP : false positive, indicating the number of positive cases that are incorrectly classified by the classifier.
- FN : false negative, indicating the number of negative cases that are incorrectly classified by the classifier.

Estimated amount	yes	no	Total
Actual amount			
yes	TP	FN	P
no	FP	TN	N
Total	P'	N'	$P + N$

Machine Learning

Measurement	Ratio
Accuracy and recognition rate	$\frac{TP + TN}{P + N}$
Error rate and misclassification rate	$\frac{FP + FN}{P + N}$
Sensitivity, true positive rate, and recall	$\frac{TP}{P}$
Specificity and true negative rate	$\frac{TN}{N}$
Precision	$\frac{TP}{TP + FP}$
F_1 , harmonic mean of the recall rate and precision	$\frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$
F_β , where β is a non-negative real number	$\frac{(1 + \beta^2) \times \text{precision} \times \text{recall}}{\beta^2 \times \text{precision} + \text{recall}}$

Machine Learning

9. Assignments Chapter 2

1. Structured representation of raw input data to meaningful ____ is called a model.

1. pattern
2. data
3. object
4. none of the above

2. For supervised learning we have ____ model.

1. interactive
2. predictive
3. descriptive
4. prescriptive

3. For unsupervised learning we have ____ model.

1. interactive
2. predictive
3. descriptive
4. prescriptive

4. Which of the following measure is not used for a classification model?

1. Accuracy
2. Recall
3. Purity
4. Error rate

Machine Learning

5. Which of the following is a performance measure for regression?

1. Accuracy
2. Recall
3. RMSE
4. Error rate

6. Which of the following is the measure of cluster quality?

Purity

2. Distance
3. Accuracy
4. all of the above

7. Out of 200 emails, a classification model correctly predicted 150 spam emails and 30 ham emails. What is the accuracy of the model?

1. 10%
2. 90%
3. 80%
4. none of the above

8. Out of 200 emails, a classification model correctly predicted 150 spam emails and 30 ham emails. What is the error rate of the model?

1. 10%
2. 90%
3. 80%
4. none of the above

Machine Learning

10. SHORT ANSWER-TYPE QUESTIONS

1. What is the main purpose of a descriptive model? State some real-world problems solved using descriptive models

2. Differentiate (any two):

1. Predictive vs. descriptive models
2. Model underfitting vs. overfitting

EPNU - 23411015
2026/03/13 13:00

Machine Learning

Chapter Three

Supervised Learning: Classification

Let us now delve into some common classification algorithms. Following are the most common classification algorithms. We will cover details of the other algorithms in this chapter

1. *k*-Nearest Neighbour (*k*NN)
2. Decision tree
3. Random forest
4. Support Vector Machine (SVM)
5. Naïve Bayes classifier

1. K Nearest Neighbors

K nearest neighbors is a simple algorithm that stores all available cases and classifies new cases based on a similarity measure (e.g., distance functions). KNN has been used in statistical estimation and pattern recognition already in the beginning of 1970's as a non-parametric technique.

Machine Learning

Algorithm

A case is classified by a majority vote of its neighbors, with the case being assigned to the class most common amongst its K nearest neighbors measured by a distance function.

If $K = 1$, then the case is simply assigned to the class of its nearest neighbor.

Distance functions

Euclidean

$$\sqrt{\sum_{i=1}^k (x_i - y_i)^2}$$

Manhattan

$$\sum_{i=1}^k |x_i - y_i|$$

Minkowski

$$\left(\sum_{i=1}^k (|x_i - y_i|)^q \right)^{1/q}$$

Machine Learning

It should also be noted that all three distance measures are only valid for continuous variables. In the instance of categorical variables the Hamming distance must be used. It also brings up the issue of standardization of the numerical variables between 0 and 1 when there is a mixture of numerical and categorical variables in the dataset.

Choosing the optimal value for K is best done by first inspecting the data. In general, a large K value is more precise as it reduces the overall noise but there is no guarantee. Cross-validation is another way to retrospectively determine a good K value by using an independent dataset to validate the K value. Historically, the optimal K for most datasets has been between 3-10. That produces much better results than 1NN.

Example:

Consider the following data concerning credit default. Age and Loan are two numerical variables (predictors) and Default is the target.

Machine Learning

Age	Loan	Default	Distance
25	\$40,000	N	102000
35	\$60,000	N	82000
45	\$80,000	N	62000
20	\$20,000	N	122000
35	\$120,000	N	22000
52	\$18,000	N	124000
23	\$95,000	Y	47000
40	\$62,000	Y	80000
60	\$100,000	Y	42000
48	\$220,000	Y	78000
33	\$150,000	Y	8000
48	\$142,000	?	

Euclidean Distance

$$D = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

We can now use the training set to classify an unknown case

(Age=48 and Loan=\$142,000) using Euclidean distance. If K=1 then the nearest neighbor is the last case in the training set with Default=Y.

$$D = \text{Sqrt}[(48-33)^2 + (142000-150000)^2] = 8000.01 \gg \text{Default=Y}$$

With K=3, there are two Default=Y and one Default=N out of three closest neighbors.

The prediction for the unknown case is again Default=Y.

Standardized Distance

One major drawback in calculating distance measures directly from the training set is in the case where variables have different measurement scales or there is a mixture of numerical and categorical variables. For example, if one variable is based on annual

Machine Learning

income in dollars, and the other is based on age in years then income will have a much higher influence on the distance calculated. One solution is to standardize the training set as shown below.

Age	Loan	Default	Distance
0.125	0.11	N	0.7652
0.375	0.21	N	0.5200
0.625	0.31	N	0.3167
0	0.01	N	0.9245
0.375	0.50	N	0.3428
0.8	0.00	N	0.6220
0.075	0.38	Y	0.6669
0.5	0.22	Y	0.4437
1	0.41	Y	0.3650
0.7	1.00	Y	0.3861
0.325	0.65	Y	0.3771
0.7	0.61	?	

Standardized Variable

$$X_s = \frac{X - Min}{Max - Min}$$

Using the standardized distance on the same training set, the unknown case returned a different neighbor which is not a good sign of robustness.

Machine Learning

2. Decision trees

A decision tree is a tree-like structure that is used as a model for classifying data. A decision tree decomposes the data into sub-trees made of other sub-trees and/or leaf nodes.

A decision tree is made up of three types of nodes

- *Decision Nodes*: These type of node have two or more branches
 - *Leaf Nodes*: The lowest nodes which represents decision
 - *Root Node*: This is also a decision node but at the topmost level
- **Entropy** is calculated based on proportion of Target Values. And formula is as follow

$$\text{Entropy} = \sum_{i=1}^T -p_i \log(p_i)$$

i is Level of Target Variable

Machine Learning

How to we build a decision tree?

Example

Consider the table below. It represent factors that affect whether John would go out to play golf or not. Using the data in the table, build a a decision tree to model that can be used to predict if John would play golf or not.



Outlook	Temperature	Humidity	Windy	Play Golf
Rainy	Hot	High	FALSE	No
Rainy	Hot	High	TRUE	No
Overcast	Hot	High	FALSE	Yes
Sunny	Mild	High	FALSE	Yes
Sunny	Cool	Normal	FALSE	Yes
Sunny	Cool	Normal	TRUE	No
Overcast	Cool	Normal	TRUE	Yes
Rainy	Mild	High	FALSE	No
Rainy	Cool	Normal	FALSE	Yes
Sunny	Mild	Normal	FALSE	Yes
Rainy	Mild	Normal	TRUE	Yes
Overcast	Mild	High	TRUE	Yes
Overcast	Hot	Normal	FALSE	Yes
Sunny	Mild	High	TRUE	No

Figure 1: Exercise on Decision Trees

Machine Learning

Solution

➤ Step 1: Determine the Decision Column

Since decision trees are used for classification, you need to determine the classes which are the basis for the decision. In this case, it is the last column, that is *Play Golf* column with classes *Yes* and *No*.

To determine the rootNode we need to compute the entropy. To do this, we create a frequency table for the classes (the Yes/No column).

Play Golf(14)	
Yes	No
9	5

Table 1: Frequency Table

Step 2: Calculating Entropy for the classes (Play Golf)

In this step, you need to calculate the entropy for the Play Golf column and the calculation step is given below.

$$Entropy(PlayGolf) = E(5,9)$$

Machine Learning

$$E(\text{PlayGolf}) = E(5,9)$$

$$= -\left(\frac{9}{14} \log_2 \frac{9}{14}\right) - \left(\frac{5}{14} \log_2 \frac{5}{14}\right)$$

$$= -(0.357 \log_2 0.357) - (0.643 \log_2 0.643)$$

$$= 0.94$$

➤ Step 3: Calculate Entropy for Other Attributes After Split

For the other four attributes, we need to calculate the entropy after each of the split.

- $E(\text{PlayGolf}, \text{Outlook})$
- $E(\text{PlayGolf}, \text{Temperature})$
- $E(\text{PlayGolf}, \text{Humidity})$
- $E(\text{PlayGolf}, \text{Windy})$

The entropy for two variables is calculated using the formula.

(don't worry if about this formula, its really easy doing the calculation

$$\text{Entropy}(S, T) = \sum_{c \in T} P(c) E(c)$$

There to calculate $E(\text{PlayGolf}, \text{Outlook})$, we would use the formula below:

$$E(\text{PlayGolf}, \text{Outlook}) = P(\text{Sunny})E(\text{Sunny}) + P(\text{Overcast})E(\text{Overcast}) + P(\text{Rainy})E(\text{Rainy})$$

Machine Learning

1- This frequency table is given below:

		PlayGolf(14)		
		Yes	No	
Outlook	Sunny	3	2	5
	Overcast	4	0	4
	Rainy	2	3	5

Table 3: Frequency Table for Outlook

the same as:

$$E(\text{PlayGolf}, \text{Outlook}) = P(\text{Sunny}) E(3,2) + P(\text{Overcast}) E(4,0) + P(\text{Rainy}) E(2,3)$$

This formula may look unfriendly, but it is quite clear. The easiest way to approach this calculation is to create a frequency table for the two variables, that is PlayGolf and Outlook.

Using the above table, we can then calculate $E(\text{PlayGolf}, \text{Outlook})$, which would then be given by the formula below

$$E(\text{PlayGolf}, \text{Outlook}) = \frac{5}{14} E(3,2) + \frac{4}{14} E(4,0) + \frac{5}{14} E(2,3)$$

Let's go ahead to calculate $E(3,2)$

We would not need to calculate the second and the third terms! This is because $E(4,0) = 0$

Machine Learning

$$E(2,3) = E(3,2)$$

$$\begin{aligned} E(\text{Sunny}) &= E(3,2) \\ &= -\left(\frac{3}{5} \log_2 \frac{3}{5}\right) - \left(\frac{2}{5} \log_2 \frac{2}{5}\right) \\ &= -(0.60 \log_2 0.60) - (0.40 \log_2 0.40) \\ &= -(0.60 * 0.737) - (0.40 * 0.529) \\ &= \mathbf{0.971} \end{aligned}$$

Just for clarification, let's show the the calculation steps

The calculation steps for $E(4,0)$:

$$\begin{aligned} E(\text{Overcast}) &= E(4,0) \\ &= -\left(\frac{4}{4} \log_2 \frac{4}{4}\right) - \left(\frac{0}{4} \log_2 \frac{0}{4}\right) \\ &= -(0) - (0) \\ &= \mathbf{0} \end{aligned}$$

The calculation step for $E(2,3)$ is given below

Machine Learning

$$\begin{aligned}E(\text{Rainy}) &= E(2,3) \\&= -\left(\frac{2}{5} \log_2 \frac{2}{5}\right) - \left(\frac{3}{5} \log_2 \frac{3}{5}\right) \\&= -(0.40 \log_2 0.40) - (0.6 \log_2 0.60) \\&= \mathbf{0.971}\end{aligned}$$

Time to put it all together.

We go ahead to calculate the $E(\text{PlayGolf}, \text{Outlook})$ by substituting the values we calculated from $E(\text{Sunny})$, $E(\text{Overcast})$ and $E(\text{Rainy})$ in the equation:

$$E(\text{PlayGolf}, \text{Outlook}) = P(\text{Sunny}) E(3,2) + P(\text{Overcast}) E(4,0) + P(\text{rainy}) E(2,3)$$

$$\begin{aligned}E(\text{PlayGolf}, \text{Outlook}) &= \frac{5}{14} E(3,2) + \frac{4}{14} E(4,0) + \frac{5}{14} E(2,3) \\&= \frac{5}{14} 0.971 + \frac{4}{14} 0.0 + \frac{5}{14} 0.971 \\&= 0.357 * 0.971 + 0.0 + 0.357 * 0.971 \\&= \mathbf{0.693}\end{aligned}$$

Machine Learning

➤ *E(PlayGolf, Temperature) Calculation*

Just like in the previous calculation, the calculation of $E(\text{PlayGolf}, \text{Temperature})$ is given below.

It is easier to do if you form the frequency table for the split for Temperature as shown.

		PlayGolf(14)		
		Yes	No	
Temperature	Hot	2	2	4
	Cold	3	1	4
	Mild	4	2	6

Table 4: Frequency Table for Temperature

$$E(\text{PlayGolf}, \text{Temperature}) = P(\text{Hot}) E(2,2) + P(\text{Cold}) E(3,1) + P(\text{Mild}) E(4,2)$$

➤ *E(PlayGolf, Humidity) Calculation*

Machine Learning

Just like in the previous calculation, the calculation of $E(\text{PlayGolf}, \text{Humidity})$ is given below. It

$$E(\text{PlayGolf}, \text{Temperature}) = 4/14 * E(\text{Hot}) + 4/14 * E(\text{Cold}) + 6/14 * E(\text{Mild})$$

$$E(\text{PlayGolf}, \text{Temperature}) = 4/14 * E(2, 2) + 4/14 * E(3, 1) + 6/14 * E(4, 2)$$

$$\begin{aligned} E(\text{PlayGolf}, \text{Temperature}) &= 4/14 * -(2/4 \log 2/4) - (2/4 \log 2/4) \\ &+ 4/14 * -(3/4 \log 3/4) - (1/4 \log 1/4) \\ &+ 6/14 * -(4/6 \log 4/6) - (2/6 \log 2/6) \end{aligned}$$

$$\begin{aligned} E(\text{PlayGolf}, \text{Temperature}) &= 5/14 * 1.0 \\ &+ 4/14 * 1.811 \\ &+ 5/14 * 0.918 \\ &= 0.911 \end{aligned}$$

It is easier to do if you form the frequency table for the split for Humidity as shown.

		PlayGolf(14)		
		Yes	No	
Humidity	High	3	4	7
	Normal	6	1	7

Table 5: Frequency Table for Humidity

Machine Learning

$$E(\text{PlayGolf}, \text{Humidity}) = 7/14 * E(\text{High}) + 7/14 * E(\text{Normal})$$

$$E(\text{PlayGolf}, \text{Humidity}) = 7/14 * E(3, 2) + 7/14 * E(4, 0)$$

$$\begin{aligned} E(\text{PlayGolf}, \text{Humidity}) &= 7/14 * -(3/7 \log 3/7) - (4/7 \log 4/7) \\ &\quad + 7/14 * -(6/7 \log 6/7) - (1/7 \log 1/7) \end{aligned}$$

$$\begin{aligned} E(\text{PlayGolf}, \text{Humidity}) &= 7/14 * 0.985 \\ &\quad + 7/14 * 0.592 \\ &= 0.788 \end{aligned}$$

➤ *E(PlayGolf, Windy) Calculation*

Just like in the previous calculation, the calculation of $E(\text{PlayGolf}, \text{Windy})$ is given below. It

It is easier to do if you form the frequency table for the split for Windy as shown.

Machine Learning

		PlayGolf(14)		
		Yes	No	
Windy	TRUE	3	3	6
	FALSE	6	2	8

Table 6: Frequency Table for Windy

$$E(\text{PlayGolf}, \text{Windy}) = 6/14 * E(\text{True}) + 8/14 * E(\text{False})$$

$$E(\text{PlayGolf}, \text{Windy}) = 6/14 * E(3, 3) + 8/14 * E(6, 2)$$

$$\begin{aligned} E(\text{PlayGolf}, \text{Windy}) &= 6/14 * -(3/6 \log 3/6) - (3/6 \log 3/6) \\ &+ 8/14 * -(6/8 \log 6/8) - (2/8 \log 2/8) \end{aligned}$$

$$\begin{aligned} E(\text{PlayGolf}, \text{Windy}) &= 6/14 * 1.0 \\ &+ 8/14 * 0.811 \\ &= 0.892 \end{aligned}$$

Machine Learning

Wow! That is so much work! So take break, walk around a little and take a glass of cold water.

Then we continue.

So now that we have all the entropies for all the four attributes, let's go ahead to summarize them as shown in below:

1. $E(\text{PlayGolf}, \text{Outlook}) = \mathbf{0.693}$
2. $E(\text{PlayGolf}, \text{Temperature}) = \mathbf{0.911}$
3. $E(\text{PlayGolf}, \text{Humidity}) = \mathbf{0.788}$
4. $E(\text{PlayGolf}, \text{Windy}) = \mathbf{0.892}$

➤ Step 4: Calculating Information Gain for Each Split

The next step is to calculate the information gain for each of the attributes. The information gain is calculated from the split using each of the attributes. Then the attribute with the largest information gain is used for the split.

The information gain is calculated using the formula:

$$\text{Gain}(S, T) = \text{Entropy}(S) - \text{Entropy}(S, T)$$

Machine Learning

For example, the information gain after splitting using the Outlook attribute is given by:

$$Gain(PlayGolf, Outlook) = Entropy(PlayGolf) - Entropy(PlayGolf, Outlook)$$

$$\begin{aligned} Gain(PlayGolf, Outlook) &= Entropy(PlayGolf) - Entropy(PlayGolf, Outlook) \\ &= 0.94 - 0.693 = \mathbf{0.247} \end{aligned}$$

$$\begin{aligned} Gain(PlayGolf, Temperature) &= Entropy(PlayGolf) - Entropy(PlayGolf, \\ Temperature) \end{aligned}$$

$$= 0.94 - 0.911 = \mathbf{0.029}$$

$$\begin{aligned} Gain(PlayGolf, Humidity) &= Entropy(PlayGolf) - Entropy(PlayGolf, Humidity) \\ &= 0.94 - 0.788 = \mathbf{0.152} \end{aligned}$$

$$\begin{aligned} Gain(PlayGolf, Windy) &= Entropy(PlayGolf) - Entropy(PlayGolf, Windy) \\ &= 0.94 - 0.892 = \mathbf{0.048} \end{aligned}$$

Having calculated all the information gain, we now choose the attribute that gives the highest information gain after the split.

Machine Learning

➤ Step 5: Perform the First Split

Draw the First Split of the Decision Tree

Now that we have all the information gain, we then split the tree based on the attribute with the highest information gain.

From our calculation, the highest information gain comes from Outlook. Therefore the split will look like this:

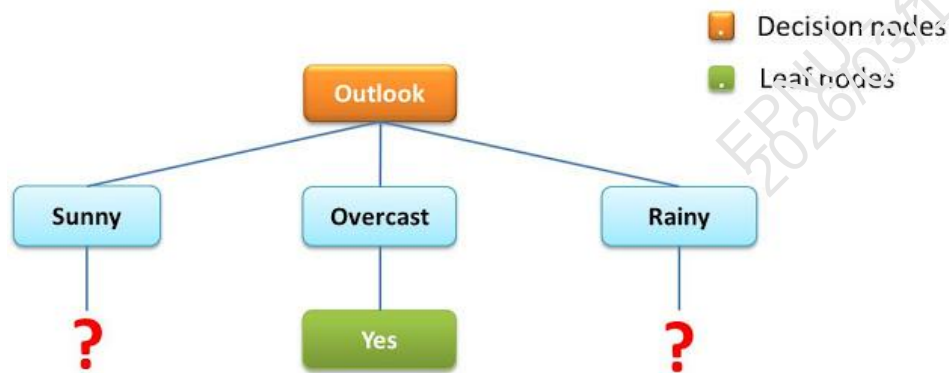


Figure 2: Decision Tree after first split

Now that we have the first stage of the decision tree, we see that we have one leaf node. But we still need to split the tree further.

To do that, we need to also split the original table to create sub tables. This sub tables are given in below.

Machine Learning

Outlook	Temperature	Humidity	Windy	Play Golf
Sunny	Mild	Normal	FALSE	Yes
Sunny	Mild	High	FALSE	Yes
Sunny	Cool	Normal	FALSE	Yes
Sunny	Cool	Normal	TRUE	No
Sunny	Mild	High	TRUE	No
Overcast	Hot	High	FALSE	Yes
Overcast	Mild	High	TRUE	Yes
Overcast	Hot	Normal	FALSE	Yes
Overcast	Cool	Normal	TRUE	Yes
Rainy	Hot	High	FALSE	No
Rainy	Hot	High	TRUE	No
Rainy	Mild	High	FALSE	No
Rainy	Cool	Normal	FALSE	Yes
Rainy	Mild	Normal	TRUE	Yes

Table 7: Initial Split using Outlook

From Table 3, we could see that the Overcast outlook requires no further split because it is just one homogeneous group. So we have a leaf node.

➤ Step 6: Perform Further Splits

The Sunny and the Rainy attributes needs to be split

The Rainy outlook can be split using either Temperature, Humidity or Windy.

Machine Learning

Example 1: What attribute would best be used for this split? Why?

Answer: **Humidity**. Because it produces homogenous groups.

Outlook	Temperature	Humidity	Windy	Play Golf
Rainy	Hot	High	FALSE	No
Rainy	Hot	High	TRUE	No
Rainy	Mild	High	FALSE	No
Rainy	Cool	Normal	FALSE	Yes
Rainy	Mild	Normal	TRUE	Yes

Table 8: Split using Humidity

The Rainy attribute could be split using High and Normal attributes and that would give us the tree below.

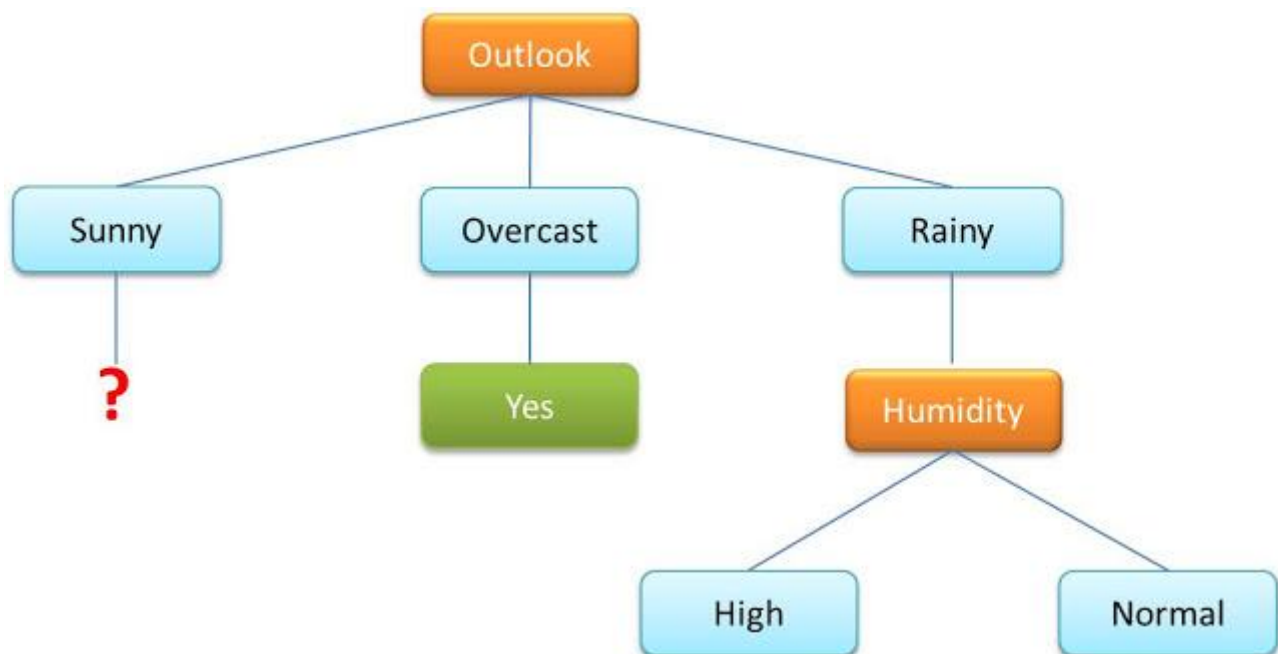


Figure 3: Split using the Humidity Attribute

Machine Learning

Let's now go ahead to do the same thing for the Sunny outlook. The Rainy outlook can be split using either Temperature, Humidity or Windy.

Quiz 2: What attribute would best be used for this split? Why?

Answer: Windy . Because it produces homogeneous groups.

Outlook	Temperature	Humidity	Windy	Play Golf
Sunny	Mild	Normal	FALSE	Yes
Sunny	Mild	High	FALSE	Yes
Sunny	Cool	Normal	FALSE	Yes
Sunny	Cool	Normal	TRUE	No
Sunny	Mild	High	TRUE	No

Table 9: Split using Windy Attribute

If we do the split using the Windy attribute, we would have the final tree that would require no further splitting! This is shown in Figure 4

➤ Step 7: Complete the Decision Tree

The complete table is shown in Figure 4

Machine Learning

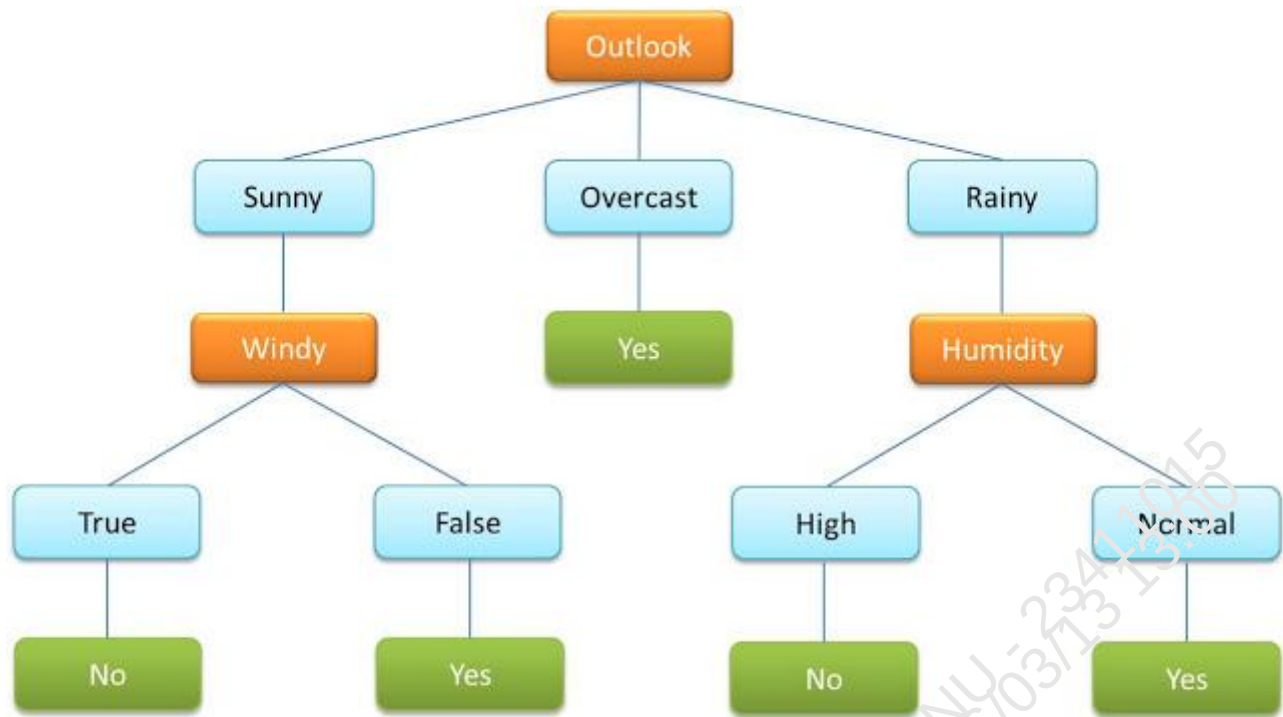
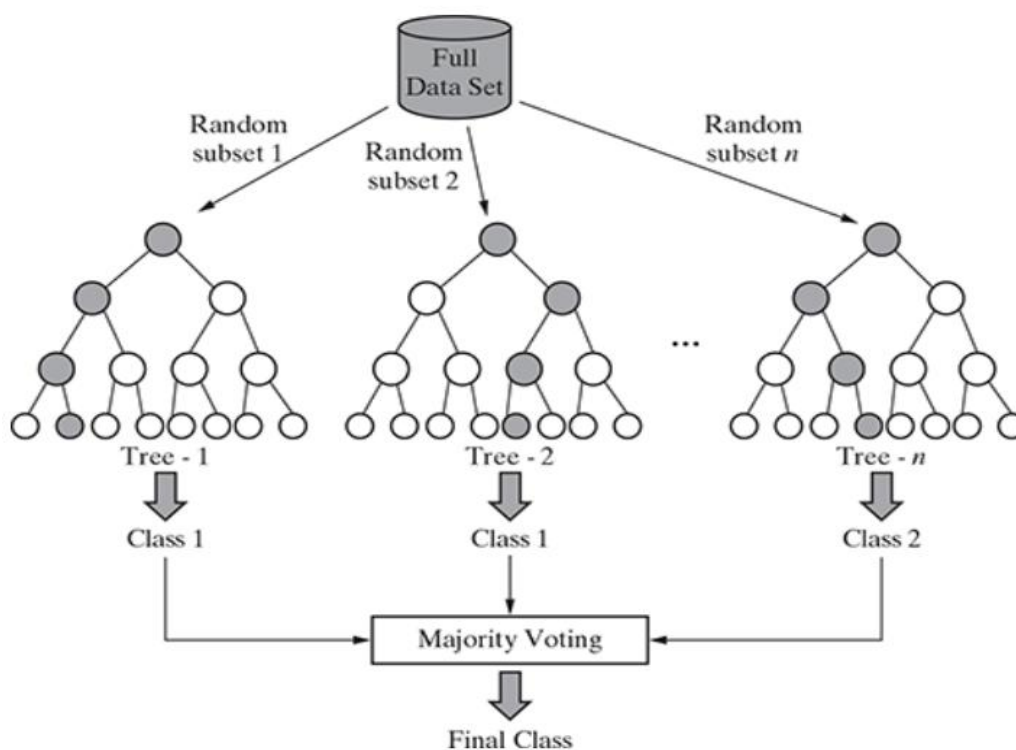


Figure 4: Final Decision Tree

Machine Learning

3. Random forest model

Random forest is an **ensemble classifier**, i.e. a combining classifier that uses and combines many decision tree classifiers. Ensembling is usually done using the concept of bagging with different feature sets. The reason for using large number of trees in random forest is to train the trees enough such that contribution from each feature comes in a number of models. After the random forest is generated by combining the trees, majority vote is applied to combine the output of the different trees. A simplified random forest model is depicted in Figure. The result from the ensemble model is usually better than that from the individual decision tree models.



Machine Learning

How does random forest work?

The random forest algorithm works as follows:

1. If there are N variables or features in the input data set, select a subset of ' m ' ($m < N$) features at random out of the N features. Also, the observations or data instances should be picked randomly.
2. Use the best split principle on these ' m ' features to calculate the number of nodes ' d '.
3. Keep splitting the nodes to child nodes till the tree is grown to the maximum possible extent.
4. Select a different subset of the training data 'with replacement' to train another decision tree following steps (1) to (3). Repeat this to build and train ' n ' decision trees.
5. Final class assignment is done on the basis of the majority votes from the ' n ' trees.

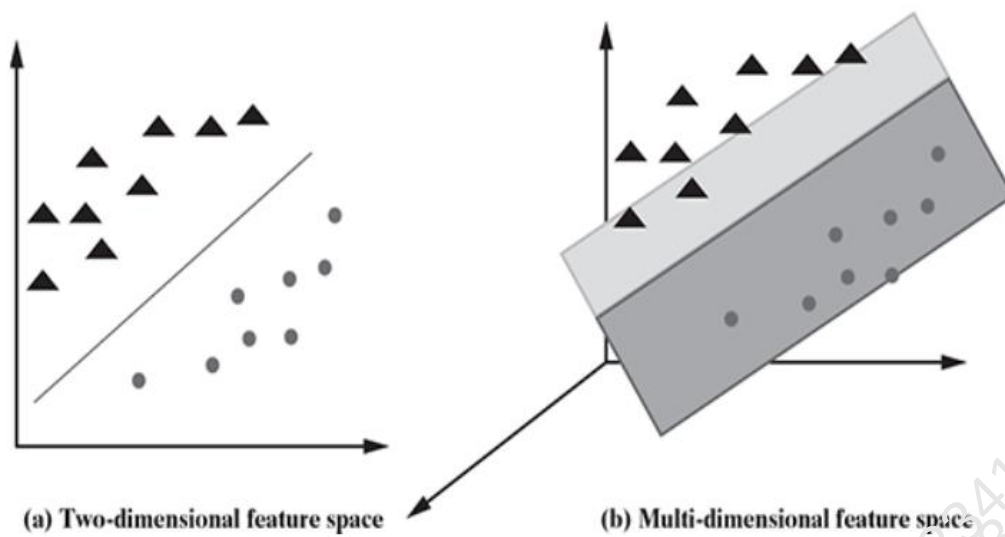
Machine Learning

4. Support vector machines

SVM is a model, which can do linear classification as well as regression. SVM is based on the concept of a surface, called a hyperplane, which draws a boundary between data instances plotted in the multi-dimensional feature space. **The output prediction of an SVM is one of two conceivable classes** which are already defined in the training data. In summary, the SVM algorithm builds an N-dimensional hyperplane model that assigns future instances into one of the two possible output classes

In SVM, a model is built to discriminate the data instances belonging to different classes. Let us assume for the sake of simplicity that the data instances are linearly separable. In this case, when mapped in a two-dimensional space, the data instances belonging to different classes fall in different sides of a straight line drawn in the two-dimensional space as depicted in Figure. If the same concept is extended to a multidimensional feature space, the straight line dividing data instances belonging to different classes transforms to a hyperplane as depicted in Figure. In other words, the **goal of the SVM analysis** is to find a plane, or rather a hyperplane, which separates the instances on the basis of their classes.

Machine Learning



Support Vectors: Support vectors are the data points (representing classes), the critical component in a data set, which are near the identified set of lines (hyperplane).

If support vectors are removed, they will alter the position of the dividing hyperplane.

Hyperplane and Margin: For an N -dimensional feature space, hyperplane is a flat subspace of dimension $(N-1)$ that separates and classifies a set of data. For example, if we consider a two-dimensional feature space (which is nothing but a data set having two features and a class variable), a hyperplane will be a one-dimensional subspace or a straight line. In the same way, for a three-dimensional feature space (data set having three features and a class variable), hyperplane is a two-dimensional subspace or a simple plane. However, quite understandably, it is difficult to visualize a feature space greater than three dimensions, much like for a subspace or hyperplane having more than three dimensions.

Machine Learning

Mathematically, in a two-dimensional space, hyperplane can be defined by the equation:

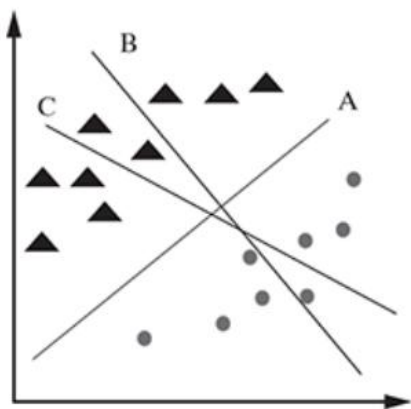
$c_0 + c_1X_1 + c_2X_2 = 0$, which is nothing but an equation of a straight line.

Identifying the correct hyperplane in SVM

As we have already discussed, there may be multiple options for hyperplanes dividing the data instances belonging to the different classes. We need to identify which one will result in the best classification. Let us examine a few scenarios before arriving to that conclusion. For the sake of simplicity of visualization, the hyperplanes have been shown as straight lines in most of the diagrams.

Scenario 1

As depicted in Figure , in this scenario, we have three hyperplanes: A, B, and C. Now, we need to identify the correct hyperplane which better segregates the two classes represented by the triangles and circles. As we can see, hyperplane 'A' has performed this task quite well.



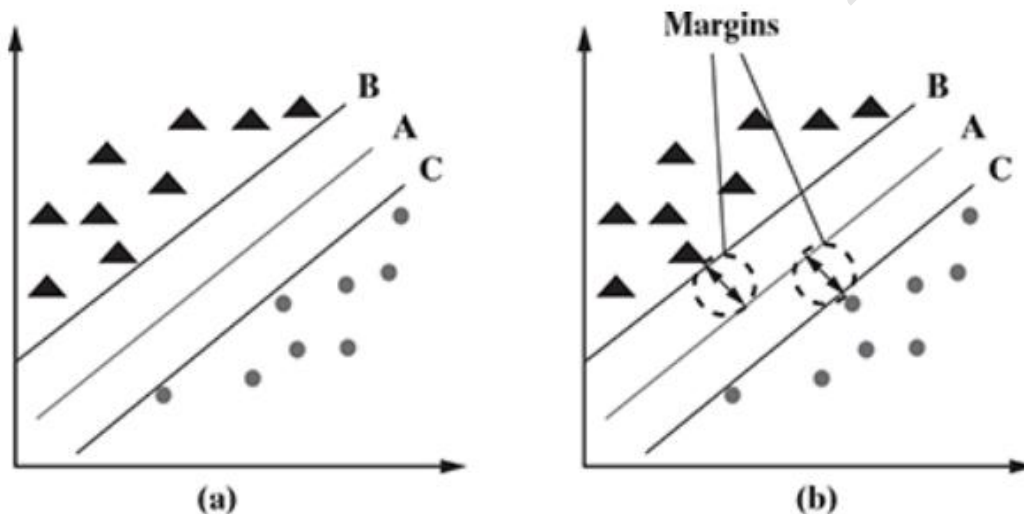
Support vector machine: Scenario 1

Machine Learning

Scenario 2

As depicted in Figure 7.17, we have three hyperplanes: A, B, and C. We have to identify the correct hyperplane which classifies the triangles and circles in the best possible way. Here, maximizing the distances between the nearest data points of both the classes and hyperplane will help us decide the correct hyperplane. This distance is called as **margin**.

In Figure , you can see that the margin for hyperplane A is high as compared to those for both B and C. Hence, hyperplane A is the correct hyperplane. Another quick reason for selecting the hyperplane with higher margin (distance) is robustness. If we select a hyperplane having a lower margin (distance), then there is a high probability of misclassification.

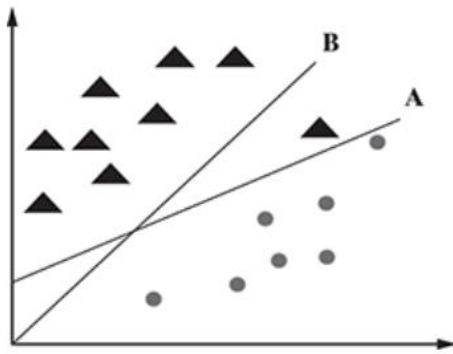


Support vector machine: Scenario 2

Machine Learning

Scenario 3

Use the rules as discussed in the previous section to identify the correct hyperplane in the scenario shown in Figure . Some of you might have selected hyperplane B as it has a higher margin (distance from the class) than A. But, here is the catch; SVM selects the hyperplane which classifies the classes accurately before maximizing the margin. Here, hyperplane B has a classification error, and A has classified all data instances correctly. Therefore, A is the correct hyperplane.



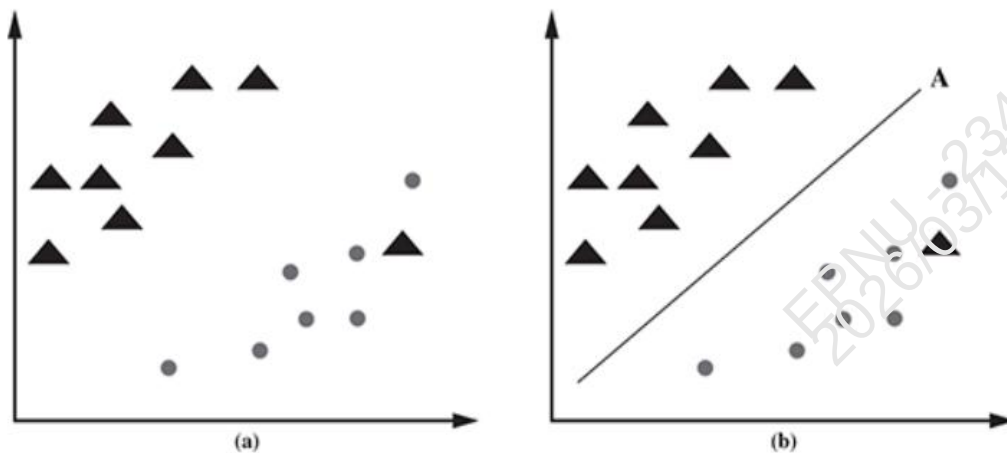
Support vector machine: Scenario 3

Machine Learning

Scenario 4

In this scenario, as shown in Figure , it is not possible to distinctly segregate the two classes by using a straight line, as one data instance belonging to one of the classes (triangle) lies in the territory of the other class (circle) as an outlier.

One triangle at the other end is like an outlier for the triangle class. SVM has a feature to ignore outliers and find the hyperplane that has the maximum margin (hyperplane A, as shown in Fig.). Hence, we can say that SVM is robust to outliers.



Support vector machine: Scenario 4

Machine Learning

5. Assignments Chapter Three

1. Predicting whether a tumour is malignant or benign is an example of?

1. Unsupervised Learning
2. Supervised Regression Problem
3. Supervised Classification Problem
4. Categorical Attribute

2. Price prediction in the domain of real estate is an example of?

1. Unsupervised Learning
2. Supervised Regression Problem
3. Supervised Classification Problem
4. Categorical Attribute

3. Let us consider two examples, say ‘predicting whether a tumour is malignant or benign’ and ‘price prediction in the domain of real estate’. These two problems are same in nature.

1. TRUE
2. FALSE

4. Supervised machine learning is as good as the data used to train it.

1. TRUE
2. FALSE

5. Which is a type of machine learning where a target feature, which is of categorical type, is predicted for the test data on the basis of the information imparted by the training data?

1. Unsupervised Learning
2. Supervised Regression
3. Supervised Classification
4. Categorical Attribute

Machine Learning

6. Classification is a type of supervised learning where a target feature, which is of categorical type, is predicted for the test data on the basis of the information imparted by the training data. The target categorical feature is known as?

1. Object
2. Variable
3. Method
4. Class

7. This is the first step in the supervised learning model.

1. Problem Identification
2. Identification of Required Data
3. Data Pre-processing
4. Definition of Training Data Set

8. This is the cleaning/transforming the data set in the supervised learning model.

1. Problem Identification
2. Identification of Required Data
3. Data Pre-processing
4. Definition of Training Data Set

9. This refers to the transformations applied to the identified data before feeding the same into the algorithm.

1. Problem Identification
2. Identification of Required Data
3. Data Pre-processing
4. Definition of Training Data Set

Machine Learning

10. This step of supervised learning determines ‘the type of training data set’.

1. Problem Identification
2. Identification of Required Data
3. Data Pre-processing
4. Definition of Training Data Set

11. Entire design of the programme is done over here in supervised learning.

1. Problem Identification
2. Training
3. Data Pre-processing
4. Definition of Training Data Set

12. Training data run on the algorithm is called as?

1. Program
2. Training
3. Training Information
4. Learned Function

13. SVM is an example of?

1. Linear Classifier and Maximum Margin Classifier
2. Non-linear Classifier and Maximum Margin Classifier
3. Linear Classifier and Minimum Margin Classifier
4. Non-linear Classifier and Minimum Margin Classifier

Machine Learning

14. ----- in terms of SVM means that an SVM is inflexible in classification

1. Hard Margin
2. Soft Margin
3. Linear Margin
4. Non-linear Classifier

15. ----- are the data points (representing classes), the important component in a data set, which are near the identified set of lines (hyperplane).

1. Hard Margin
2. Soft Margin
3. Linear Margin
4. Support Vectors

16. ----- is a line that linearly separates and classifies a set of data.

1. Hyperplane
2. Soft Margin
3. Linear Margin
4. Support Vectors

17. The distance between hyperplane and data points is called as:

1. Hyper Plan
2. Margins
3. Error
4. Support Vectors

Machine Learning

18. Which of the following is true about SVM?

1. It is useful only in high-dimensional spaces
2. It always gives an approximate value
3. It is accurate
4. Understanding SVM is difficult

19. Which of the following is true about SVM?

1. It is useful only in high-dimensional spaces
2. It requires less memory
3. SVM does not perform well when we have a large data set
4. SVM performs well when we have a large data set

20. What is the meaning of hard margin in SVM?

1. SVM allows very low error in classification
2. SVM allows high amount of error in classification
3. Underfitting
4. SVM is highly flexible

21. What sizes of training data sets are not best suited for SVM?

1. Large data sets
2. Very small training data sets
3. Medium size training data sets
4. Training data set size does not matter

22. Support Vectors are near the hyperplane.

1. True
2. False

Machine Learning

23. In SVM, these functions take a lower dimensional input space and transform it to a higher dimensional space.

1. Kernels
2. Vector
3. Support Vector
4. Hyperplane

24. Which of the following options is true about the k NN algorithm?

1. It can be used only for classification
2. It can be used only for regression
3. It can be used for both classification and regression
4. It is not possible to use for both classification and regression

25. Which of the following will be Euclidean distance between the two data points A(4,3) and B(2,3)?

1. 1
2. 2
3. 4
4. 8

26. Which of the following will be Manhattan distance between the two data points A(8,3) and B(4,3)?

1. 1
2. 2
3. 4
4. 8

Machine Learning

27. When you find many noises in data, which of the following options would you consider in k NN?

1. Increase the value of k
2. Decrease the value of k
3. Noise does not depend on k
4. $K = 0$

28. What would be the relationship between the training time taken by 1-NN, 2-NN, and 3-NN?

1. $1\text{-NN} > 2\text{-NN} > 3\text{-NN}$
2. $1\text{-NN} < 2\text{-NN} < 3\text{-NN}$
3. $1\text{-NN} \sim 2\text{-NN} \sim 3\text{-NN}$
4. None of these

29. Which of the following algorithms is an example of the ensemble learning algorithm?

1. Random Forest
2. Decision Tree
3. k NN
4. SVM

30. Which of the following is not an inductive bias in a decision tree?

1. It prefers longer tree over shorter tree
2. Trees that place nodes near the root with high information gain are preferred
3. Overfitting is a natural phenomenon in a decision tree
4. Prefer the shortest hypothesis that fits the data

Machine Learning

6. SHORT ANSWER-TYPE QUESTIONS

1. What is supervised learning? Why it is called so?
2. Give an example of supervised learning in a hospital industry.
3. Give any three examples of supervised learning.
4. What is classification and regression in a supervised learning?
5. Give some examples of common classification algorithms.
6. Explain, in brief, the SVM model.
7. What is cost of misclassification in SVM?
8. Define Support Vectors in the SVM model.
9. Define kernel in the SVM model

EPNU - 23411015
2026/03/13 13:00

Machine Learning

Chapter Four

Bayesian Concept Learning

Bayesian learning algorithms, like the naive Bayes classifier, are highly practical approaches to certain types of learning problems as they can calculate explicit probabilities for hypotheses. In many cases, they are equally competitive or even outperform the other learning algorithms, including decision tree and neural network algorithms.

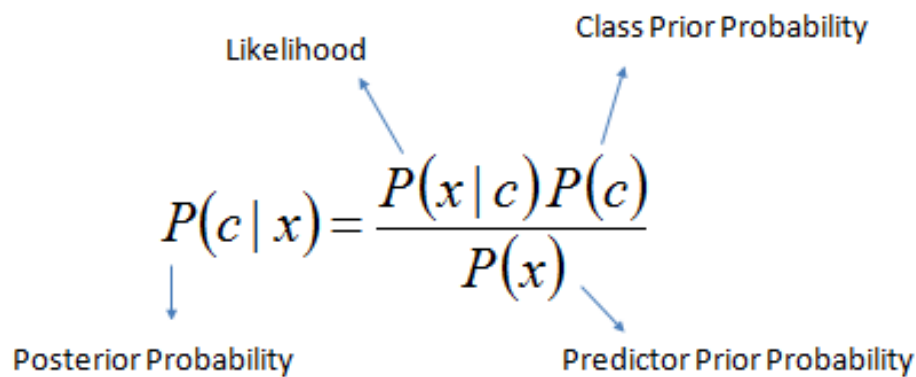
1. Naive Bayes Classification

The Naive Bayesian classifier is based on Bayes' theorem with the independence assumptions between predictors. A Naive Bayesian model is easy to build, with no complicated iterative parameter estimation which makes it particularly useful for very large datasets. Despite its simplicity, the Naive Bayesian classifier often does surprisingly well and is widely used because it often outperforms more sophisticated classification methods.

Algorithm

Bayes theorem provides a way of calculating the posterior probability, $P(c|x)$, from $P(c)$, $P(x)$, and $P(x|c)$. Naive Bayes classifier assume that the effect of the value of a predictor (x) on a given class (c) is independent of the values of other predictors. This assumption is called class conditional independence.

Machine Learning



The diagram shows the Bayes' theorem formula: $P(c | x) = \frac{P(x | c)P(c)}{P(x)}$. Arrows point from labels to the corresponding parts of the formula: 'Likelihood' points to $P(x | c)$, 'Class Prior Probability' points to $P(c)$, 'Posterior Probability' points to $P(c | x)$, and 'Predictor Prior Probability' points to $P(x)$.

$$P(c | X) = P(x_1 | c) \times P(x_2 | c) \times \dots \times P(x_n | c) \times P(c)$$

- $P(c|x)$ is the posterior probability of *class (target)* given *predictor (attribute)*.
- $P(c)$ is the prior probability of *class*.
- $P(x|c)$ is the likelihood which is the probability of *predictor* given *class*.
- $P(x)$ is the prior probability of *predictor*.

In ZeroR model there is no predictor, in OneR model we try to find the single best predictor, naive Bayesian includes all predictors using Bayes' rule and the independence assumptions between predictors.

Machine Learning

Example 1:

We use the same simple Weather dataset here.

Outlook	Temp	Humidity	Windy	Play Golf
Rainy	Hot	High	False	No
Rainy	Hot	High	True	No
Overcast	Hot	High	False	Yes
Sunny	Mild	High	False	Yes
Sunny	Cool	Normal	False	Yes
Sunny	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Rainy	Mild	High	False	No
Rainy	Cool	Normal	False	Yes
Sunny	Mild	Normal	False	Yes
Rainy	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Sunny	Mild	High	True	No

The posterior probability can be calculated by first, constructing a frequency table for each attribute against the target. Then, transforming the frequency tables to likelihood tables and finally use the Naive Bayesian equation to calculate the posterior probability for each class.

The class with the highest posterior probability is the outcome of prediction.

Machine Learning

$$P(x | c) = P(\text{Sunny} | \text{Yes}) = 3 / 9 = 0.33$$

Frequency Table		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3



Likelihood Table		Play Golf		
		Yes	No	
Outlook	Sunny	3/9	2/5	5/14
	Overcast	4/9	0/5	4/14
	Rainy	2/9	3/5	5/14
		9/14	5/14	

$$P(x) = P(\text{Sunny}) = 5 / 14 = 0.36$$

$$P(c) = P(\text{Yes}) = 9 / 14 = 0.64$$

Posterior Probability:

$$P(c | x) = P(\text{Yes} | \text{Sunny}) = 0.33 \times 0.64 \div 0.36 = 0.60$$



EPNU - 23411015
2026/03/13 13:00

$$P(x | c) = P(\text{Sunny} | \text{No}) = 2 / 5 = 0.4$$

Frequency Table		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3



		Play Golf		
		Yes	No	
Outlook	Sunny	3	2	5
	Overcast	4	0	4
	Rainy	2	3	5
		9	5	14

$$P(x) = P(\text{Sunny}) = 5 / 14 = 0.36$$

$$P(c) = P(\text{No}) = 5 / 14 = 0.36$$

Posterior Probability:

$$P(c | x) = P(\text{No} | \text{Sunny}) = 0.40 \times 0.36 \div 0.36 = 0.40$$



Machine Learning

The likelihood tables for all four predictors.

Frequency Table

		Play Golf	
		Yes	No
Outlook	Sunny	3	2
	Overcast	4	0
	Rainy	2	3



Likelihood Table

		Play Golf	
		Yes	No
Outlook	Sunny	3/9	2/5
	Overcast	4/9	0/5
	Rainy	2/9	3/5

		Play Golf	
		Yes	No
Humidity	High	3	4
	Normal	6	1



		Play Golf	
		Yes	No
Humidity	High	3/9	4/5
	Normal	6/9	1/5

		Play Golf	
		Yes	No
Temp.	Hot	2	2
	Mild	4	2
	Cool	3	1



		Play Golf	
		Yes	No
Temp.	Hot	2/9	2/5
	Mild	4/9	2/5
	Cool	3/9	1/5

		Play Golf	
		Yes	No
Windy	False	6	2
	True	3	3



		Play Golf	
		Yes	No
Windy	False	6/9	2/5
	True	3/9	3/5

Machine Learning

Example 2:

In this example we have 4 inputs (predictors). The final posterior probabilities can be standardized between 0 and 1.

Outlook	Temp	Humidity	Windy	Play
Rainy	Cool	High	True	?

$$P(Yes | X) = P(Rainy | Yes) \times P(Cool | Yes) \times P(High | Yes) \times P(True | Yes) \times P(Yes)$$

$$P(Yes | X) = 2/9 \times 3/9 \times 3/9 \times 3/9 \times 9/14 = 0.00529 \rightarrow 0.2 = \frac{0.00529}{0.02057 + 0.00529}$$

$$P(No | X) = P(Rainy | No) \times P(Cool | No) \times P(High | No) \times P(True | No) \times P(No)$$

$$P(No | X) = 3/5 \times 1/5 \times 4/5 \times 3/5 \times 5/14 = 0.02057 \rightarrow 0.8 = \frac{0.02057}{0.02057 + 0.00529}$$

Machine Learning

2. SUMMARY

- ✚ Bayesian methods introduced a basis for probabilistic learning methods that consider the knowledge about the **prior** probabilities of alternative hypotheses and the probability of **likelihood** or observing various training data given the hypothesis. It then assigns a **posterior** probability to each candidate hypothesis on the basis of the assumed priors and the observed data.

EPNU - 23411015
2026/03/13 13:00

Machine Learning

3. Assignments Chapter four

1. Three companies X, Y, and Z supply 40%, 45%, and 15% of the uniforms to a school. Past experience shows that 2%, 3%, and 4% of the uniforms supplied by these companies are defective. If a uniform was found to be defective, what is the probability that the uniform was supplied by Company X?

1. $\frac{34}{69}$

2. $\frac{25}{55}$

3. $\frac{16}{55}$

4. $\frac{16}{12}$

EPNU - 23411015
2026/03/13 13:00

Machine Learning

Chapter Five

Supervised Learning: Regression

1. Introduction

In the last **three chapters**, you have got quite a good conceptual overview of supervised learning algorithm for categorical data prediction. You got a detailed understanding of all the popular models of classification that are used by machine learning practitioners to solve a wide array of prediction problems where the target variable is a categorical variable.

In this chapter, we will build concepts on **prediction of numerical variables** – which is another key area of supervised learning. This area, known as **regression**, focuses on solving problems such as predicting value of real estate, demand forecast in retail, weather forecast, etc

In the context of regression, dependent variable (Y) is the one whose value is to be predicted, e.g. the price quote of the real estate in the context of Karen's problem. This variable is presumed to be functionally related to one (say, X) or more independent variables called predictors. In the context of Karen's problem, Frank used area of the property, location, floor, etc. as predictors of the model that he built. In other words, the dependent variable depends on independent variable(s) or predictor(s). Regression is essentially finding a relationship (or) association between

Machine Learning

the dependent variable (Y) and the independent variable(s) (X), i.e. to find the function ' f ' for the association $Y = f(X)$.

2. COMMON REGRESSION ALGORITHMS

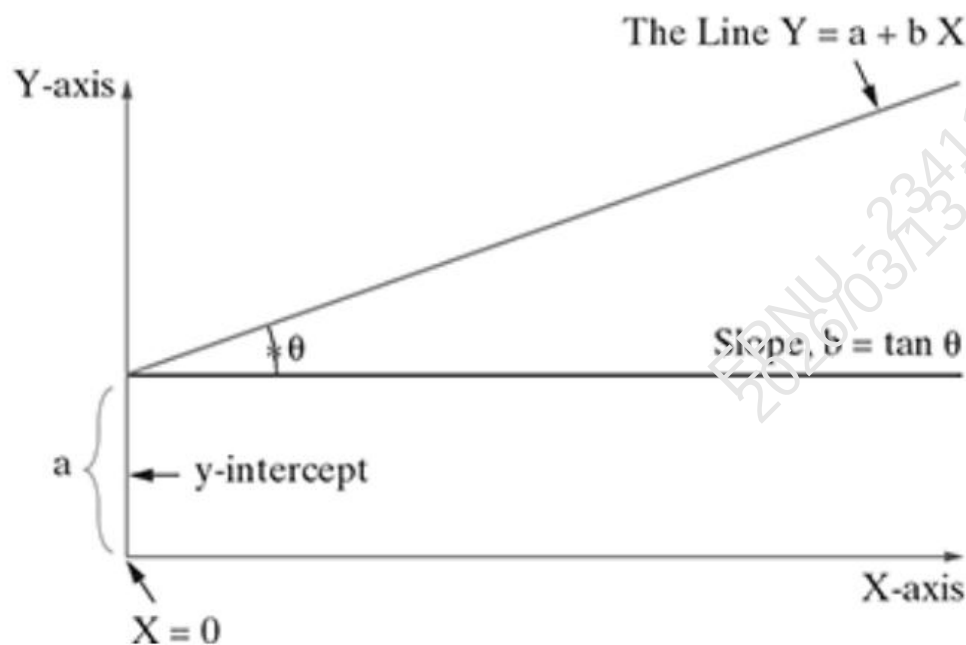
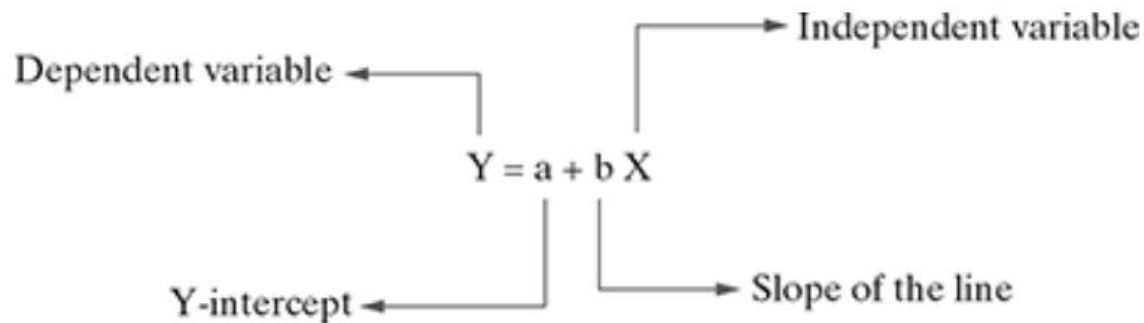
The most common regression algorithms are

- ✚ Simple linear regression
- ✚ Multiple linear regression
- ✚ Polynomial regression
- ✚ Multivariate adaptive regression splines Logistic regression
- ✚ Maximum likelihood estimation (least squares)

2.1 Simple Linear Regression

As the name indicates, simple linear regression is the simplest regression model which involves only one predictor. This model assumes a linear relationship between the dependent variable and the predictor variable as shown in Figure

Machine Learning



In the context of Karen's problem, if we take Price of a Property as the dependent variable and the Area of the Property (in sq. m.) as the predictor variable, we can build a model using simple linear regression.

$$\text{Price}_{\text{Property}} = f(\text{Area}_{\text{Property}})$$

Machine Learning

Assuming a linear association, we can reformulate the model as

$$\text{Price}_{\text{Property}} = a + b \cdot \text{Area}_{\text{Property}}$$

where ‘ a ’ and ‘ b ’ are intercept and slope of the straight line, respectively.

Just to recall, straight lines can be defined in a slope– intercept form $Y = (a + bX)$, where a = intercept and b = slope of the straight line. The value of intercept indicates the value of Y when $X = 0$. It is known as ‘the intercept or Y intercept’ because it specifies where the straight line crosses the vertical or Y -axis

2.1.1 Slope of the simple linear regression model

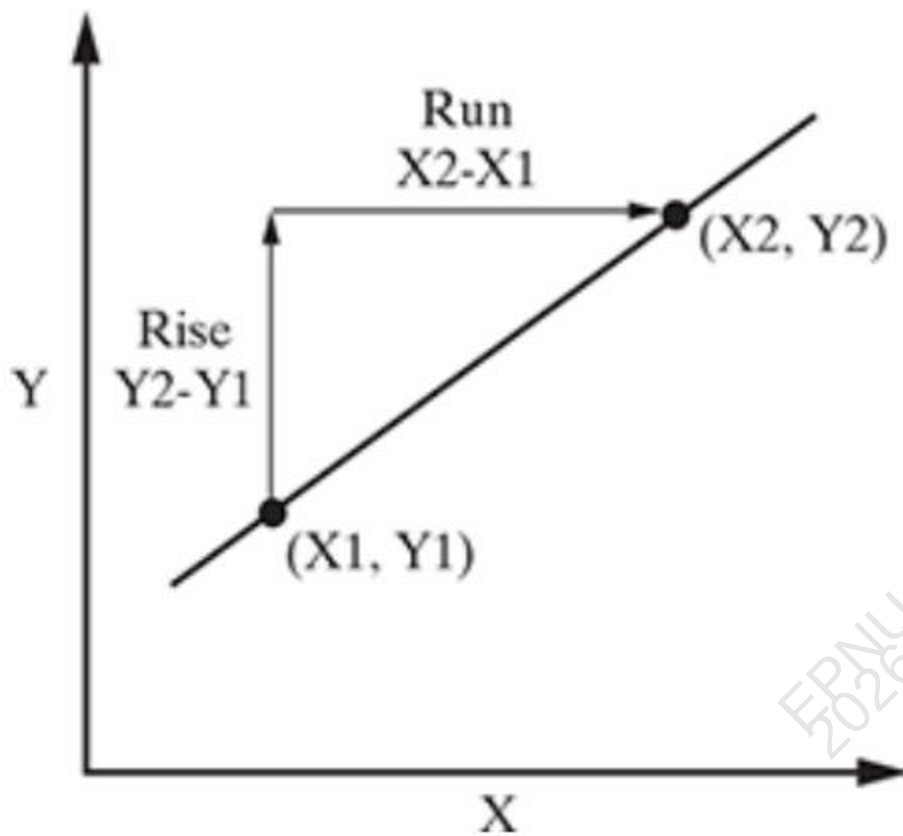
Slope of a straight line represents how much the line in a graph changes in the vertical direction (Y -axis) over a change in the horizontal direction (X -axis) as shown in Figure 8.2.

Slope = Change in Y /Change in X

Rise is the change in Y -axis ($Y - Y$) and Run is the change

in X -axis ($X - X$). So, slope is represented as given below:

Machine Learning



$$\text{Slope} = \frac{\text{Rise}}{\text{Run}} = \frac{Y_2 - Y_1}{X_2 - X_1}$$

Machine Learning

Let us find the slope of the graph where the lower point on the line is represented as $(-3, -2)$ and the higher point on the line is represented as $(2, 2)$.

$$(X, Y) = (-3, -2) \text{ and } (X, Y) = (2, 2)$$

$$\text{Rise} = (Y - Y) = (2 - (-2)) = 2 + 2 = 4$$

$$\text{Run} = (X - X) = (2 - (-3)) = 2 + 3 = 5$$

$$\text{Slope} = \text{Rise/Run} = 4/5 = 0.8$$

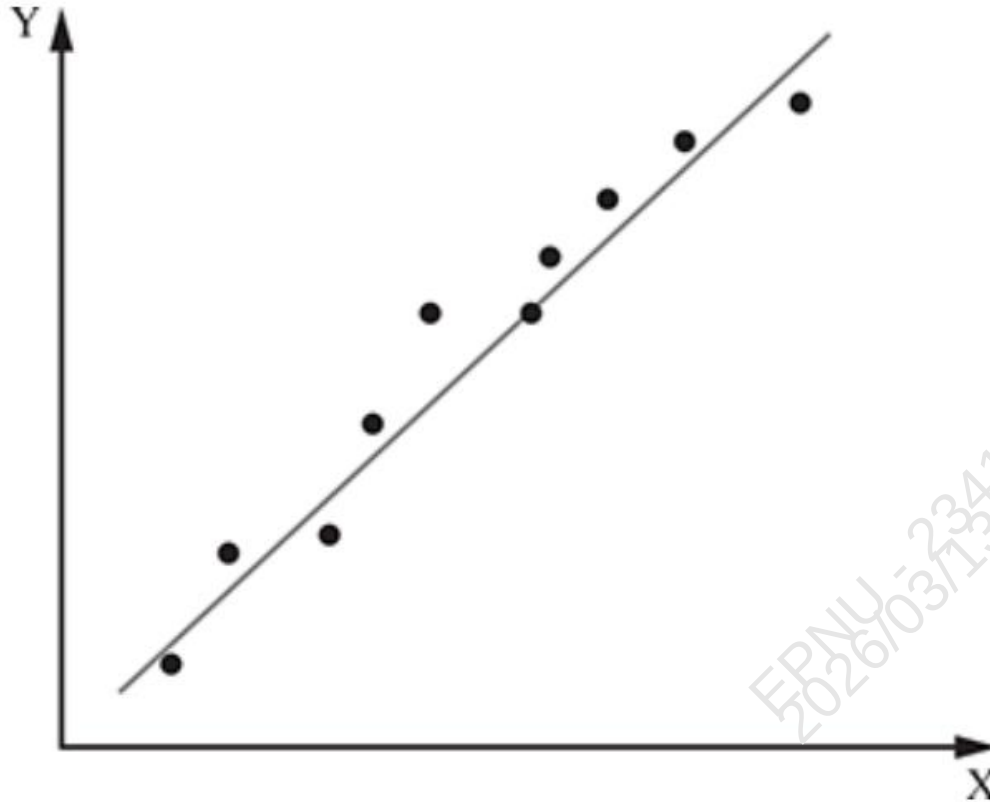
There can be two types of slopes in a linear regression model: positive slope and negative slope. Different types of regression lines based on the type of slope include

- ✚ **Linear positive slope**
- ✚ **Curve linear positive slope**
- ✚ **Linear negative slope**
- ✚ **Curve linear negative slope**

Linear positive slope

A positive slope always moves upward on a graph from left to right

Machine Learning

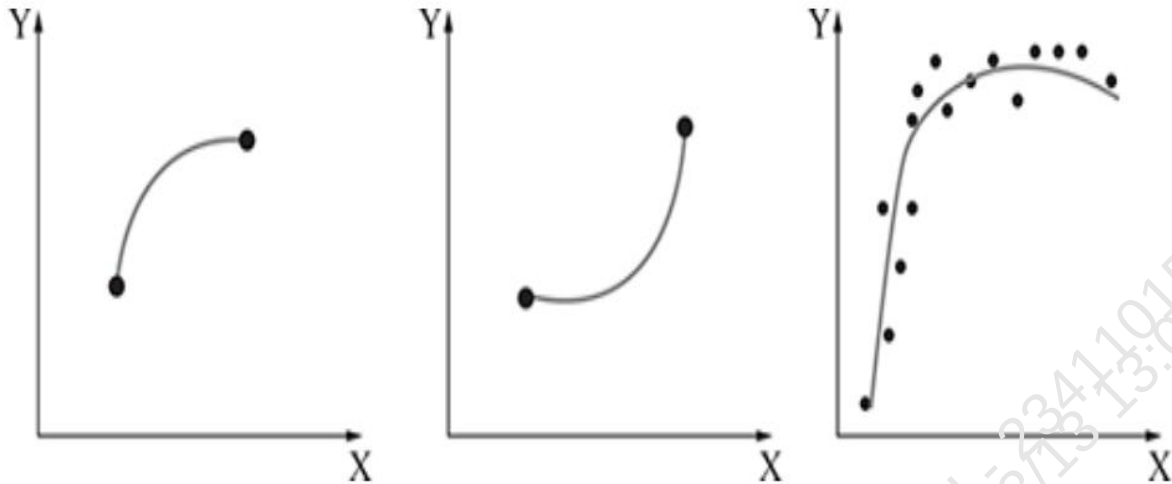


$$\text{Slope} = \text{Rise/Run} = (Y_2 - Y_1) / (X_2 - X_1) = \text{Delta } (Y) / \text{Delta}(X)$$

- ✚ Scenario 1 for positive slope: Delta (Y) is positive and Delta (X) is positive
- ✚ Scenario 2 for positive slope: Delta (Y) is negative and Delta (X) is negative

Machine Learning

Curve linear positive slope



Curves in these graphs (refer to Fig. 8.4) slope upward from left to right.

$$\text{Slope} = (Y - Y) / (X - X) = \text{Delta } (Y) / \text{Delta}(X)$$

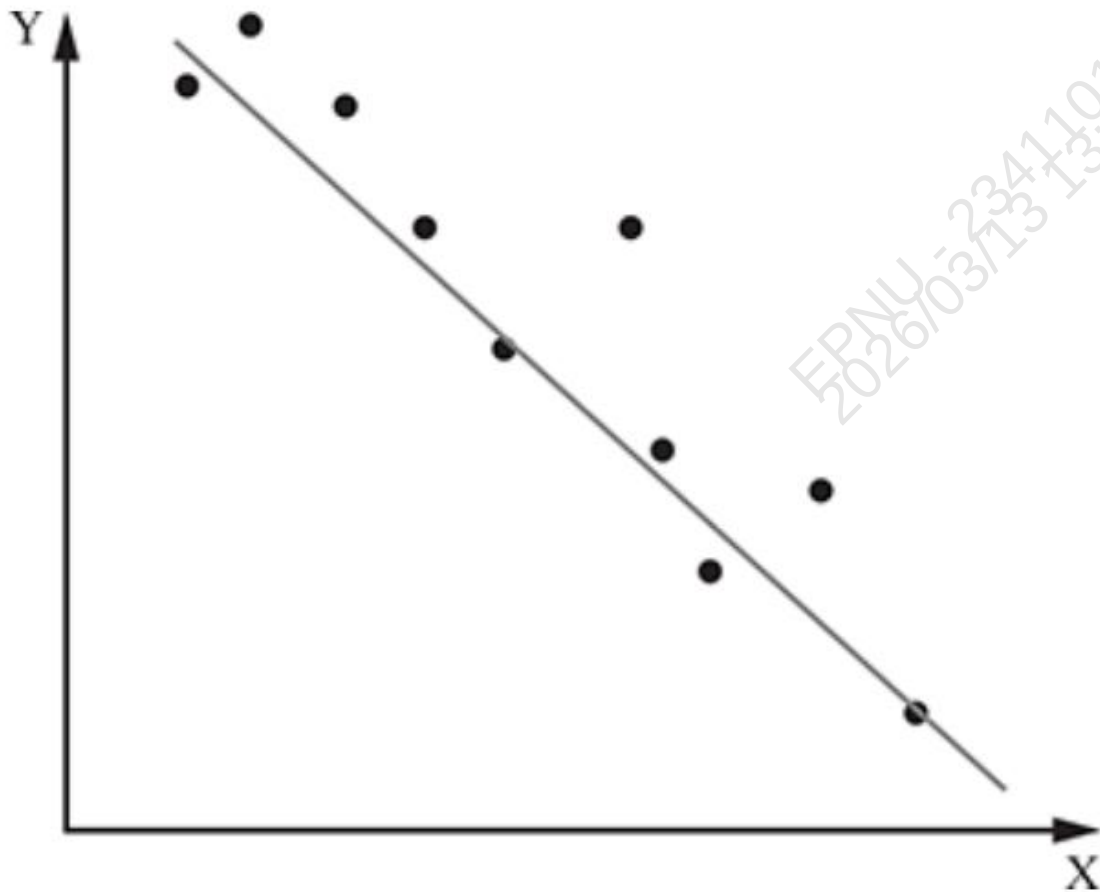
Slope for a variable (X) may vary between two graphs, but it will always be positive; hence, the above graphs are called as graphs with curve linear positive slope.

Machine Learning

Linear negative slope

A negative slope always moves downward on a graph from left to right.

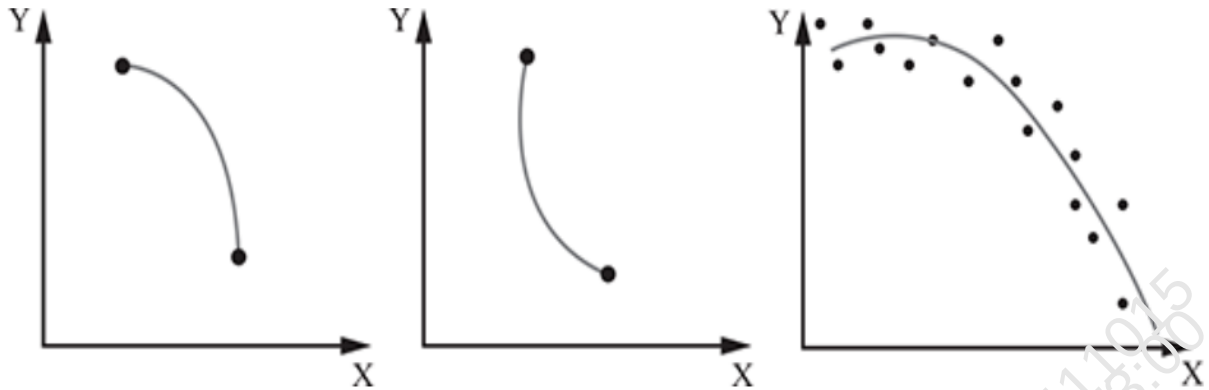
As X value (on X -axis) increases, Y value decreases



- ✚ Scenario 1 for negative slope: Delta (Y) is positive and Delta (X) is negative
- ✚ Scenario 2 for negative slope: Delta (Y) is negative and Delta (X) is positive

Machine Learning

Curve linear negative slope



Curves in these graphs slope downward from left to right.

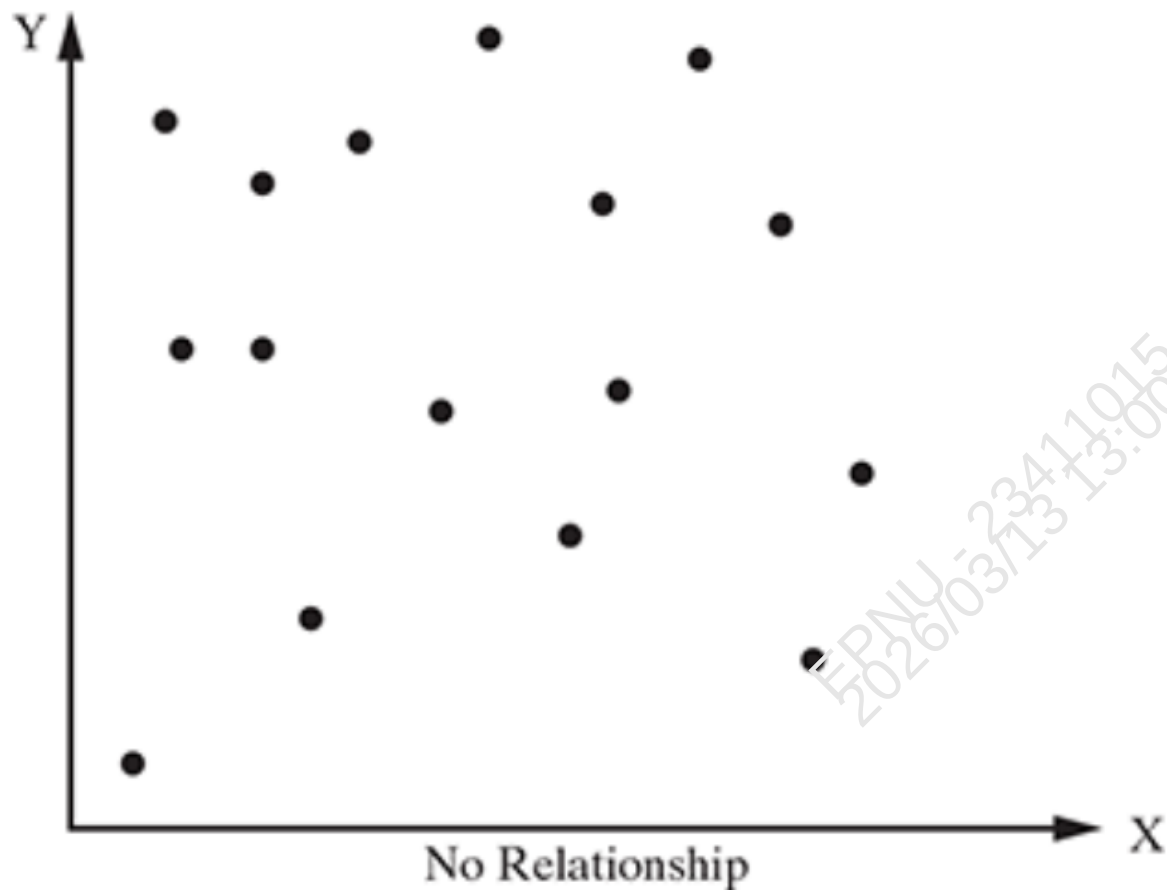
$$\text{Slope} = (Y - Y) / (X - X) = \text{Delta}(Y) / \text{Delta}(X)$$

Slope for a variable (X) may vary between two graphs, but it will always be negative; hence, the above graphs are called as graphs with curve linear negative slope.

2.1.2 No relationship graph

Scatter graph shown in Figure 8.7 indicates ‘no relationship’ curve as it is very difficult to conclude whether the relationship between X and Y is positive or negative.

Machine Learning



2.1.3 Error in simple regression

The regression equation model in machine learning uses the above slope–intercept format in algorithms. X and Y values are provided to the machine, and it identifies the values of a (intercept) and b (slope) by relating the values of X and Y. However, identifying the exact match of values for a and b is not always possible. There will be some error value (ϵ) associated with it. This error is called marginal or residual error.

Machine Learning

$$Y = (a + bX) + \varepsilon$$

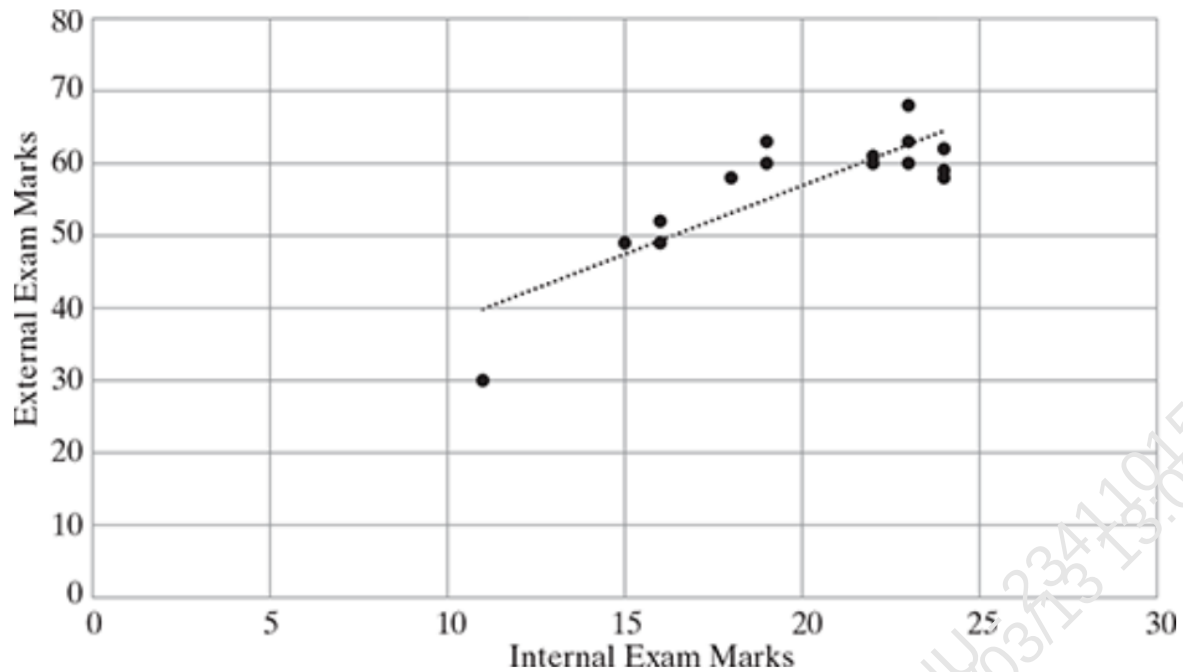
Example:

A college professor believes that if the grade for internal examination is high in a class, the grade for external examination will also be high. A random sample of 15 students in that class was selected, and the data is given below:

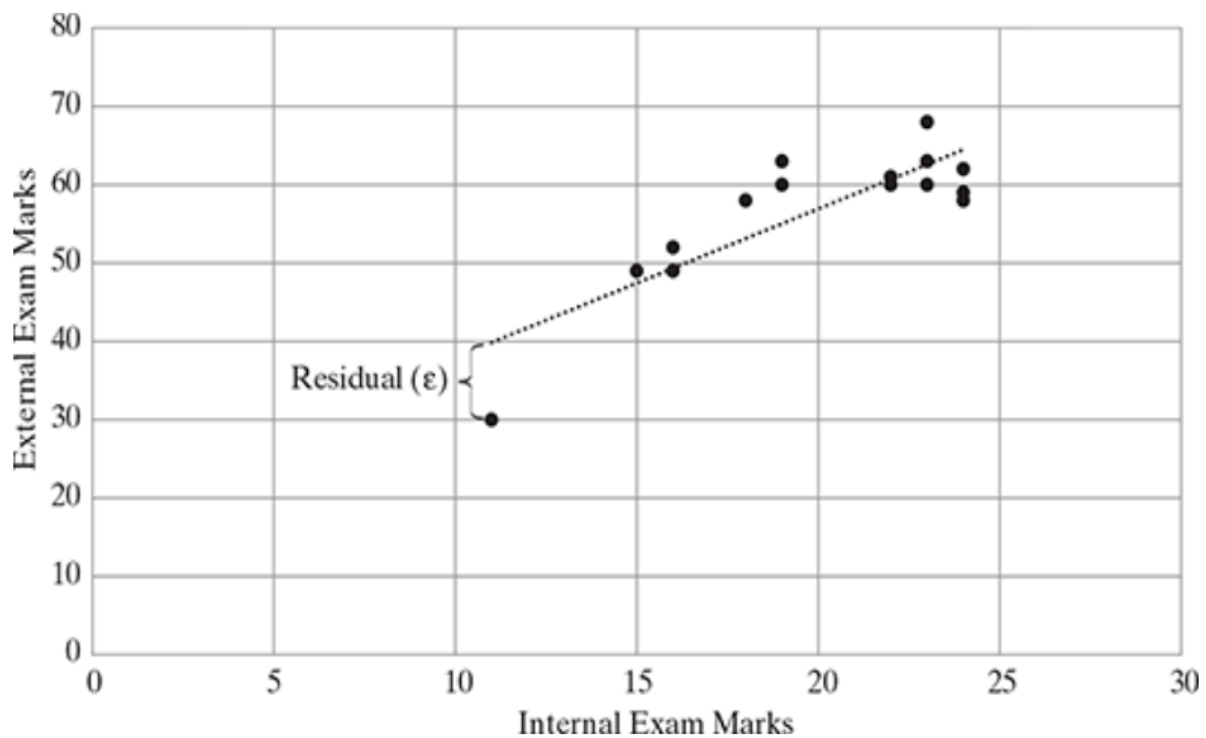
Internal Exam	15	23	18	23	24	22	22	19	19	16	24	11	24	16	23
External Exam	49	63	58	60	58	61	60	63	60	52	62	50	59	49	68

A scatter plot was drawn to explore the relationship between the independent variable (internal marks) mapped to X-axis and dependent variable (external marks) mapped to Y-axis as depicted in Figure

Machine Learning



Residual is the distance between the predicted point (on the regression line) and the actual point as depicted in Figure



Machine Learning

As we know, in simple linear regression, the line is drawn using the regression formula.

$$Y = (a + bX) + \varepsilon$$

A straight line is drawn as close as possible over the points on the scatter plot. Ordinary Least Squares (OLS) is the technique used to estimate a line that will minimize the error (ε), which is the difference between the predicted and the actual values of Y . This means summing the errors of each prediction or, more appropriately, the Sum of the Squares of

the Errors (SSE) $\left(\text{i.e. } \sum_i \varepsilon_i^2 \right)$.

It is observed that the SSE is least when b takes the value

$$b = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{\text{cov}(X, Y)}{\text{Var}(X)}$$

The corresponding value of ' a ' calculated using the above value of ' b ' is

$$a = \bar{Y} - b\bar{X}$$

Machine Learning

Calculation summary

Sum of $X = 299$

Sum of $Y = 852$

Mean X , $M_X = 19.93$

Mean Y , $M_Y = 56.8$

Sum of squares (SS_X) = 226.9333

Sum of products (SP) = 429.8

Regression equation = $\hat{y} = bX + a$

$$b = \frac{SP}{SS_X} = \frac{429.8}{226.93} = 1.89395$$

$$a = M_Y - bM_X = 56.8 - (1.89 \times 19.93) = 19.0473$$

$$\hat{y} = 1.89395X + 19.0473$$

In the context of the given problem, we can say

$$\text{Marks in external exam} = 19.04 + 1.89 \times (\text{Marks in internal exam})$$

Machine Learning

$$\text{or, } M_{\text{Ext}} = 19.04 + 1.89 \times M_{\text{Int}}$$

		Step 2		Step 3	Step 5
X	Y	X- mean (X)	Y- Mean (Y)	$(X_i - \bar{X})(Y_i - \bar{Y})$	$(X_i - \bar{X})^2$
15	49	-4.93	-7.8	38.454	24.3049
23	63	3.07	6.2	19.034	9.4249
18	58	-1.93	1.2	-2.316	3.7249
23	60	3.07	3.2	9.824	9.4249
24	58	4.07	1.2	4.884	16.5649
22	61	2.07	4.2	8.694	4.2849
22	60	2.07	3.2	6.624	4.2849
19	63	-0.93	6.2	-5.766	0.8649
19	60	-0.93	3.2	-2.976	0.8649
16	52	-3.93	-4.8	18.864	15.4449
24	62	4.07	5.2	21.164	16.5649
11	30	-8.93	-26.8	239.324	79.7449
24	59	4.07	2.2	8.954	16.5649
16	49	-3.93	-7.8	30.654	15.4449
23	68	3.07	11.2	34.384	9.4249
19.9	56.8		$\Sigma(X_i - \bar{X})(Y_i - \bar{Y})$	429.8	226.9335
Step 1		Step 4		Step 6	

Machine Learning

Step 7: Divide (step4 / step6)

$$b = 429.28 / 226.93 = 1.89$$

Step 8: Calculate a using the value of b

$$a = \bar{Y} - b\bar{X}$$

$$a = 56.8 - 1.89 \times 19.9$$

$$a = 19.05$$

2.2 Multiple Linear Regression

In a multiple regression model, two or more independent variables, i.e. predictors are involved in the model. If we think in the context of Karen's problem, in the last section, we came up with a simple linear regression by considering Price of a Property as the dependent variable and the Area of the Property (in sq. m.) as the predictor variable. However, location, floor, number of years since purchase, amenities available, etc. are also important predictors which should not be ignored. Thus, if we consider Price of a Property (in \$) as the dependent variable and Area of the Property (in sq. m.), location, floor, number of years since purchase and amenities available as the independent variables, we can form a multiple regression equation as shown below:

$$\text{Price}_{\text{Property}} = f(\text{Area}_{\text{Property}}, \text{location}, \text{floor}, \text{Ageing}, \text{Amenities})$$

Machine Learning

The simple linear regression model and the multiple regression model assume that the dependent variable is continuous.

The following expression describes the equation involving the relationship with two predictor variables, namely X_1 and X_2 .

$$\hat{Y} = a + b_1X_1 + b_2X_2$$

The model describes a plane in the three-dimensional space of $\hat{Y}$, X_1 , and X_2 . Parameter ' a ' is the intercept of this plane. Parameters ' b_1 ' and ' b_2 ' are referred to as **partial regression coefficients**. Parameter b_1 represents the change in the mean response corresponding to a unit change in X_1 when X_2 is held constant. Parameter b_2 represents the change in the mean response corresponding to a unit change in X_2 when X_1 is held constant.

Consider the following example of a multiple linear regression model with two predictor variables, namely X_1 and X_2

$$\hat{Y} = 22 + 0.3X_1 + 1.2X_2$$

Machine Learning

Multiple regression for estimating equation when there are ‘ n ’ predictor variables is as follows:

$$\hat{Y} = a + b_1X_1 + b_2X_2 + b_3X_3 + \dots + b_nX_n$$

EPNU - 23411015
2026/03/13 13:00

Machine Learning

3. Assignments chapter 5

1. When we are trying to predict a real-value variable such as ‘\$’,

‘Weight’, the problem falls under the category of

1. Unsupervised learning
2. Supervised regression problem
3. Supervised classification problem
4. Categorical attribute

2. Price prediction of crude oil is an example of

1. Unsupervised learning
2. Supervised regression problem
3. Supervised classification problem
4. Categorical attribute

3. Value to be predicted in machine learning is called as

1. Slope
2. Regression
3. Independent variable
4. Dependent variable

Machine Learning

4. This is called as predictor.

1. Slope
2. Regression
3. Independent variable
4. Dependent variable

5. This is essentially finding a relationship (or) association between the dependent variable (Y) and the independent variables (X).

1. Slope
2. Regression
3. Classification
4. Categorization

6. If the regression involves only one independent variable, it is called as

1. Multiple regression
2. One regression
3. Simple regression
4. Independent regression

7. Which equation represents simple imperfect relationship?

1. $Y = (a + bx) + \varepsilon$
2. $Y = (a + bx)$
3. $DY = \text{Change in } Y / \text{Change in } X$
4. $Y = a + b_1X_1 + b_2X_2$

Machine Learning

8. Which equation represents simple perfect relationship?

1. $Y = (a + bx) + \varepsilon$
2. $Y = (a + bx)$
3. $DY = \text{Change in } Y / \text{Change in } X$
4. $Y = a + b_1X_1 + b_2X_2$

9. What is the formula for slope in a simple linear equation?

1. $Y = (a + bx) + \varepsilon$
2. $Y = (a + bx)$
3. $DY = \text{Change in } Y / \text{Change in } X$
4. $Y = a + b_1X_1 + b_2X_2$

10. This slope always moves upward on a graph from left to right.

1. Multilinear slope
2. No relationship slope
3. Negative slope
4. Positive slope

11. This slope always moves downwards on a graph from left to right.

1. Multilinear slope
2. No relationship slope
3. Negative slope
4. Positive slope

Machine Learning

12. Maximum and minimum points on a graph are found at points where the slope of the curve is

1. Zero
2. One
3. 0.5
4. Random number

EPNU - 23411015
2026/03/13 13:00

4. SHORT ANSWER-TYPE QUESTIONS

1. What is a dependent variable and an independent variable in a linear equation?
2. What is simple linear regression? Give one example.
3. Define slope in a linear regression.
4. Find the slope of the graph where the lower point on the line is represented as $(-3, -2)$ and the higher point on the line is represented as $(2, 2)$.
5. What are the conditions of a positive slope in linear regression?
6. What are the conditions of a negative slope in linear regression?
7. What is multiple linear regression?
8. Define sum of squares due to error in multiple linear regression.
9. Define sum of squares due to regression in multiple linear regression

Machine Learning

Chapter Six

Unsupervised Learning

Unsupervised learning is a machine learning concept where the **unlabelled** and unclassified information is analysed to discover hidden knowledge. The algorithms work on the data without any prior training, but they are constructed in such a way that they can identify patterns, groupings, sorting order, and numerous other interesting knowledge from the set of data

1. Clustering

Clustering refers to a broad set of techniques for finding subgroups, or clusters, in a data set on the basis of the characteristics of the objects within that data set in such a manner that the objects within the group are similar (or related to each other) but are different from (or unrelated to) the objects from the other groups. Clustering in statistics refers to how data is gathered (“clustered”) by factors like:

- Age.
- Household size.
- Income.
- Or education level.

Sorting data into clusters sometimes leads to more investigation into the data.

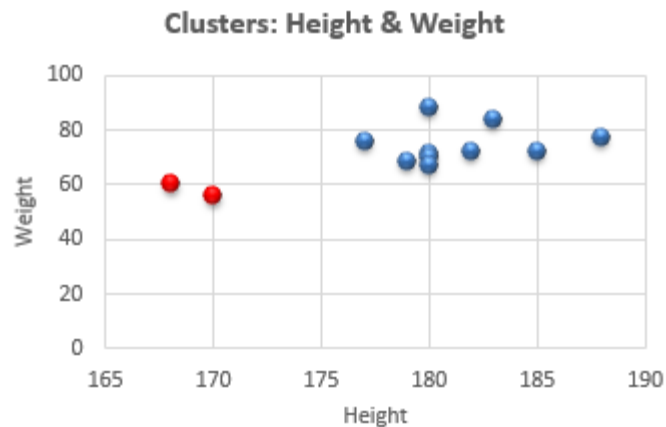
For example, cancer clusters can indicate some problem in the environment. Or, they can just be a result of the nature being random. Cluster Analysis tends to be subjective in many cases; it depends on what you perceive as common threads in the data. The technique isn't really anything new in statistics; if you've ever made a bar graph, you've probably already made clusters (even if you didn't call it that). For example, a bar graph showing breeds of dog requires you to cluster by breed (Siberian Husky,

Machine Learning

Border Collie, German Shepherd...) or a chart of income levels might be clustered by low, middle and high income levels.

Cluster analysis results showing three different colored clusters.

Business Scenario: We have height and weight information. Using these two variables, we need to group the objects based on height and weight information.



If you look at the above chart, you will expect that there are two visible clusters/segments and we want these to be identified using **K Means algorithm**.

Machine Learning

Clustering Example

Height	Weight
185	72
170	56
168	60
179	68
182	72
188	77
180	71
180	70
183	84
180	88
180	67
177	76

Step 1: Input

Dataset, Clustering Variables and Maximum Number of Clusters (K in Means Clustering)

In this dataset, only two variables –height and weight – are considered for clustering

Machine Learning

Height	Weight
185	72
170	56
168	60
179	68
182	72
188	77
180	71
180	70
183	84
180	88
180	67
177	76

Step 2: Initialize cluster centroid

In this example, value of K is considered as 2. Cluster centroids are initialized with first 2 observations.

Cluster	Initial Centroid	
	Height	Weight
K ₁	185	72
K ₂	170	56

Step 3: Calculate Euclidean Distance

Euclidean is one of the distance measures used on K Means algorithm. Euclidean distance between of a observation and initial cluster centroids 1 and 2 is calculated. Based on euclidean distance each observation is assigned to one of the clusters - based on minimum distance.

Machine Learning

$$\text{Euclidean Distance} = \sqrt{(X_H - H_1)^2 + (X_W - W_1)^2}$$

Where

X_H : Observation value of variable Height |

H_1 : Centroid value of Cluster 1 for variable Height

X_W : Observation Value of variable Weight

W_1 : Centroid value of cluster 1 for variable Weight

First two observations

Height	Weight
185	72
170	56

Now initial cluster centroids are :

Cluster	Updated Centroid	
	Height	Weight
K_1	185	72
K_2	170	56

Euclidean Distance Calculation from each of the clusters is calculated.

Euclidian Distance from Cluster 1	Euclidian Distance from Cluster 2	Assignment
$(185-185)^2 + (72-72)^2$ $= 0$	$(185-170)^2 + (72-56)^2$ $= 21.93$	1
$(170-185)^2 + (56-72)^2$ $= 21.93$	$(170-170)^2 + (56-56)^2$ $= 0$	2

Machine Learning

We have considered two observations for assignment only because we knew the assignment. And there is no change in Centroids as these two observations were only considered as initial centroids

Step 4: Move on to next observation and calculate Euclidean Distance

Height	Weight
168	60

Euclidean Distance from Cluster 1	Euclidean Distance from Cluster 2	Assignment
$(168-185)^2+(60-72)^2$ =20.808	$(168-185)^2+(60-72)^2$ = 4.472	2

Since distance is minimum from cluster 2, so the observation is assigned to cluster 2.

Machine Learning

Now revise Cluster Centroid – mean value Height and Weight as Cluster Centroids. Addition is only to cluster 2, so centroid of cluster 2 will be updated

Updated cluster centroids

Cluster	Updated Centroid	
	Height	Weight
K=1	185	72
K=2	$(170+168)/2 = 169$	$(56+60)/2 = 58$

Step 5: Calculate Euclidean Distance for the next observation, assign next observation based on minimum euclidean distance and update the cluster centroids.

Next Observation.

Height	Weight
179	68

Euclidean Distance Calculation and Assignment

Euclidean Distance from Cluster 1	Euclidean Distance from Cluster 2	Assignment
7.211103	14.14214	1

Machine Learning

Update Cluster Centroid

Cluster	Updated Centroid	
	Height	Weight
K=1	182	70.6667
K=2	169	58

Continue the steps until all observations are assigned

Final assignments

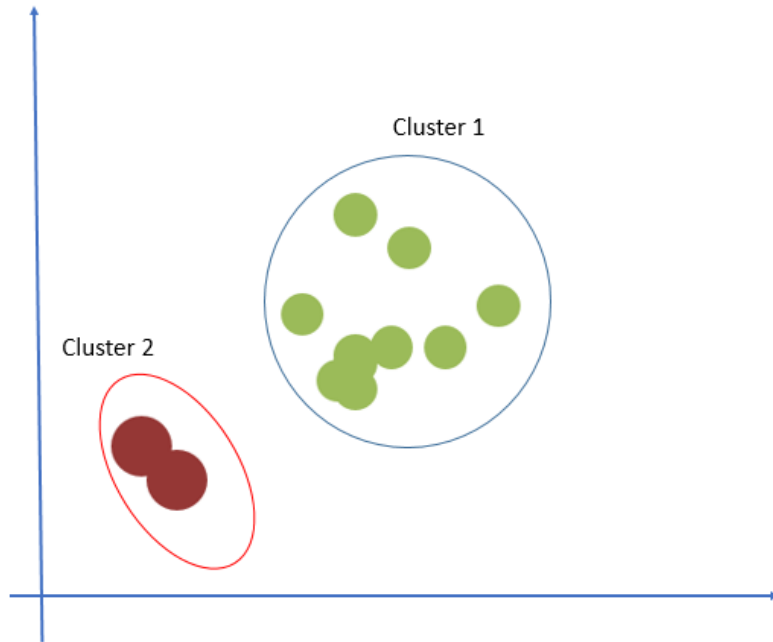
Height	Weight	Assignment
185	72	1
170	56	2
168	60	2
179	68	1
182	72	1
188	77	1
180	71	1
180	70	1
183	84	1
180	88	1
180	67	1
177	76	1

Cluster Centroids

Cluster	Updated Centroid	
	Height	Weight
K=1	182.8	72
K=2	169	58

This is what was expected initially based on two-dimensional plot.

Machine Learning



EPNU - 23411015
2026/03/13 13:00

Machine Learning

2. SUMMARY

- ✚ Unsupervised learning is a machine learning concept where the unlabelled and unclassified information is analysed to discover hidden knowledge. The algorithm works on the data without any prior training, but they are constructed in such a way that they can identify patterns, groupings, sorting order, and numerous other interesting knowledge from the set of data.
- ✚ Clustering refers to a broad set of techniques for finding subgroups, or clusters, in a data set based on the characteristics of the objects within the data set itself in such a manner that the objects within the group are similar (or related to each other) but are different from (or unrelated to) the objects from the other groups.
- ✚ k -means and k -medoids are the most popular partitioning techniques. The principle of k -means algorithm is to assign each of the n data points to one of the k clusters, where k is a user-defined parameter as the number of clusters desired. The objective is to maximize the homogeneity within the clusters and to maximize the differences between the clusters. The homogeneity and differences are measured in terms of the distance between the points.

Machine Learning

3. Assignment Chapter Six

1. One of the disadvantages of k -means algorithm is that the outliers may reduce the quality of the final clustering.

1. True
2. False

2. Which of the following can be possible termination conditions in KMeans?

1. For a fixed number of iterations.
2. Assignment of observations to clusters does not change between iterations. Except for cases with a bad local minimum.
3. Centroids do not change between successive iterations.
4. All of the above

EPNU - 23411015
2026/03/13 13:00

Machine Learning

4. SHORT ANSWER-TYPE QUESTIONS

1. How unsupervised learning is different from supervised learning?

Explain with some examples.

2. You are given a set of one-dimensional data points: {5, 10, 15, 20, 25, 30, 35}.

Assume that $k = 2$ and first set of random centroid is selected as {15, 32} and then it is refined with {12, 30}.

Create two clusters with each set of centroid mentioned above following the k -means approach

EPNU - 23411015
2026/03/13 13:00

Machine Learning

References

- ✚ Principles of Machine Learning: The Three Perspectives – *Wenmin Wang* (2024) — A modern structured guide covering theory and frameworks across supervised, unsupervised, and reinforcement learning.
- ✚ Machine Learning for Engineers – *Marcus Neuer* (2024) — Focuses on explainable ML and physics-informed methods for engineering applications.
- ✚ Machine Learning with Python: Principles and Practical Techniques – *Parteek Bhatia* (2024) — In-depth Python implementation of classic and modern ML methods.